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| | |
|---|---|
| Project Name | Predictive Temperature Monitoring Bracelet for Children |
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| Team Members / His Department | Maria Abu Alzollf – manufacturing<br>Diab Ammari – Design<br>Abdulla Aljboor – Marketing |
| HTU Course Number and Title | 102321 / Professional Engineering Management |
| Academic Year | Fall 2025 – 2026 |
| Course Tutor | Dr. Mohammad Abu-Rumman |

# Reliability In Our Project

## Introduction

Reliability remains the key to our predictive temperature-monitoring bracelet, as the device is intended for continuous use in an unpredictable real-world environment, especially at night when parents are asleep. To ensure trustworthy performance, the system integrates predictive sensing, fail-safe alert mechanisms, environmental compensation, and redundancy across both hardware and software layers.

First, the bracelet enhances reliability by shifting from a reactive to a predictive monitoring method. Instead of waiting for a fever to reach dangerous levels, the device analyzes temperature trends such as slope and acceleration—to forecast rising spikes early. This predictive logic acts as a safety margin, giving parents more time to intervene and reducing the likelihood of missing a critical event.

Secondly, the system incorporates multiple fail-safe procedures to ensure operation even when external systems fail. Unlike many smart wearables that rely heavily on Wi-Fi or smartphone connectivity, the bracelet includes offline alerts (buzzer, vibration, and LED) to guarantee that notifications are delivered regardless of network issues or app crashes.

Third, reliability is strengthened through redundancy and fault-prevention measures. The design allows several bracelets to connect to a single parent unit, minimizing the impact of individual sensor failure. Additional safeguards—such as ambient temperature compensation, battery-health alerts, placement detection, and mechanical anti-loosening components—not only reduce false alarms but also maintain stable, long-term performance.

Finally, FMEA was used to identify the most vulnerable failure modes—such as sensor drift, battery depletion, accidental triggering, and data inconsistencies—and implement corrective actions to significantly reduce their risk priority numbers.

## safety margins, fail-safe mechanisms and  process redundancies

A core reliability requirement for a child-monitoring medical wearable is the ability to operate correctly even under unpredictable or adverse conditions. The bracelet incorporates multiple fail-safe mechanisms that ensure the system continues to function even if individual components fail. For example, the device includes an **offline alerting system** (integrated buzzer and vibration motor), ensuring that parents receive real-time warnings without relying on Wi-Fi or mobile connectivity. This prevents a single point of failure, especially during nighttime power or network outages.

Safety margins are also built into the temperature-trend prediction logic. Instead of triggering alerts only when a fever threshold is exceeded, the algorithm analyzes the **rate of**

**temperature change** and **acceleration** to forecast potential spikes early. This provides a safety buffer of several minutes, allowing parents to intervene before a dangerous temperature level is reached. The design includes an **ambient temperature sensor** to compensate for environmental heat fluctuations, reducing false alarms and increasing reliability under varying room conditions.

Redundancy is supported through the ability to pair **multiple bracelets** with the same parent unit. If one bracelet fails due to battery depletion or sensor malfunction, the system maintains monitoring coverage through the remaining units. Additional reliability features include **battery-health alerts**, **placement warnings**, and **locking washers** to prevent mechanical loosening, all contributing to a robust fail-safe architecture. These mechanisms ensure dependable operation, reduce the likelihood of undetected failures, and significantly enhance overall system reliability.

## Failure Mode and Effects Analysis (FMEA)

(FMEA) is a basic method used to find possible problems before they happen.

In the inventory or warehouse area, FMEA is usually done by the team members who understand the daily operations and know where failures can happen. These people include:

- **Warehouse Manager** – oversees the process and approves actions
- **Receiving Supervisor** – identifies problems in receiving materials
- **Quality Inspector** – checks for quality-related risks
- **Dispatch Supervisor** – identifies risks in packing and shipping

These team members work together to find possible failures in receiving, storing, and dispatching materials, so they can prevent problems before they occur.

| # | Failure Mode | Effect on System / Child | Possible Cause(s) | S | O | D | RPN | Recommended Action |
|---|---|---|---|---|---|---|---|---|
| 1 | **Battery failure** (battery empty or damaged) | Bracelet stops working; no temperature monitoring; no alarm during fever | Low battery capacity, poor charging habits, battery aging, electrical fault | 9 | 4 | 3 | 108 | Use high-quality rechargeable battery, add **low-battery warning** (LED/buzzer), design for minimum 8–12 hours operation, include charging instructions for parents. |
| 2 | **Alarm failure** (buzzer or vibration does not work) | Fever spike not reported to parents; high safety risk for child | Defective buzzer/motor, software bug, broken connection, muted alarm | 10 | 3 | 4 | 120 | Add **self-test** for alarm at start-up, use redundant alert (buzzer + vibration + LED), clear alarm volume setting, periodic maintenance check. |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 3 | **Case failure** (bracelet housing cracks or opens) | Internal electronics exposed; risk of water damage or injury; device may stop working | Poor material quality, impact from falling, child pulls or bites the bracelet | 7 | 5 | 4 | 140 | Use stronger, child-safe material, rounded design with no sharp edges, drop-test during design, design secure but serviceable closure. |
| 4 | **Screen broken** (display damaged or unreadable) | Parents cannot read temperature or status; may ignore alarm or misinterpret condition | Mechanical shock, child hitting or dropping device, low impact resistance | 6 | 4 | 5 | 120 | Use **scratch- and impact-resistant** cover , slightly recessed screen, optional mobile/parent unit display as backup. |
| 5 | **Sensor failure** (temperature sensor cracked or not reading correctly) | Wrong or no temperature reading; false alarms or missed fever | Manufacturing defect, aging, moisture ingress, case damage, poor assembly | 10 | 3 | 3 | 90 | Use reliable medical-grade sensor, protect it with waterproof design, add **sensor self-diagnostic** (check for unrealistic values), calibration during production. |

### *1) Highest RPN – Case Failure (RPN = 140)*

Case failure received the highest RPN because it has a high chance of happening and also affects multiple parts of the bracelet. Since the device is worn by children, it may fall, hit objects, or be pulled or bitten. This increases the Occurrence rating. A cracked or opened case exposes the internal electronics to damage, moisture, or dust, which leads to further failures such as sensor malfunction or battery damage. Although the severity is not as high as alarm failure, the combination of a high Occurrence and medium Detection makes its RPN the highest. This means the case design must be improved, strengthened, and tested to reduce the risk.

### *2) Lowest RPN – Sensor Failure (RPN = 90)*

Sensor failure has the lowest RPN among the listed failure modes. Although the severity is high (because wrong temperature readings are dangerous), the chance of it happening (Occurrence) is relatively low when using medical-grade sensors. In addition, sensor issues can often be detected early through software checks, such as unrealistic values or missing signals, giving it a better Detection score. Because the failure is easier to detect and less likely to happen, its overall RPN becomes the lowest. This means regular calibration, waterproof sealing, and a self-diagnostic function are sufficient to keep this risk under control.

# literature review

## History of the Problem: Fever Monitoring in Children

Fever is one of the most common signs of illness in children and often serves as an early warning of infection or other medical problems. While most fevers are manageable, they can escalate quickly, especially if they are not detected early. In some cases, complications such as febrile seizures may occur when a child's temperature rises too fast. This risk is higher among newborns under three months, whose immune systems are still developing, and children with conditions such as cancer or sickle cell anemia, where fever may indicate a bacterial infection — one of the major causes of mortality in these groups **[1], [2]**.

Traditional manual thermometers have historically been used for fever detection, but they come with major limitations. Parents must physically wake up and check the child's temperature frequently, which is inconvenient at night and increases the chance of missing sudden temperature spikes. Manual measurements also depend heavily on correct placement and user attention, making them prone to human error **[1]**.

As wearable technology has grown, new fever-monitoring tools such as smart thermometers and smartwatches have been introduced. These devices can monitor temperature continuously and provide alerts, but their accuracy is still questionable. Many of them rely on sensors originally designed to measure device temperature rather than human body temperature, which leads to inaccurate readings. They are also affected by room temperature and movement, and most depend on Wi-Fi or mobile apps to deliver notifications. These systems can fail during power outages or when the parent's phone is out of range, making them unreliable during critical times,  especially at night **[3], [4], [5]**.

Recent research on non-invasive, wireless, wearable sensors shows potential for better continuous fever monitoring. However, issues such as placement, accuracy, and the delay in alerting remain challenges. This leaves a gap for a more reliable, predictive fever-monitoring solution designed specifically for children **[6]**.

## Existing solutions

Traditional thermometers are still widely used, but they require manual checking. This means parents must wake up repeatedly at night, which is not always reliable, especially since a fever can rise very quickly. If the thermometer is not positioned correctly or if the parent forgets to check, the child's fever may go unnoticed, leading to late responses and inaccurate readings.

Wearable devices like **TempTraq** and **Fever Scout** monitor temperature continuously, but these tools still have serious limitations. Notifications are often delayed until the temperature has already reached a dangerous level, leaving little time for the parent to react. Many of these devices also rely heavily on Wi-Fi or phone apps, which can disconnect or fail when the battery dies or during power cuts. This makes them less dependable in nighttime conditions when early alerts matter the most.

Smartwatches and fitness bands have also been suggested for fever monitoring, but they primarily measure **skin temperature**, not actual core body temperature. Skin readings are influenced by room temperature, physical activity, and clothing, making them less accurate. Similar to other devices, they usually depend on apps or connectivity that may stop working during emergencies.

*Comparison Table*

| Solution | Accuracy | Real-time Alerts | Works Offline | Predictive | Comfort |
|---|---|---|---|---|---|
| **Manual Thermometer** | Low | No | Yes | No | High |
| **TempTraq / Fever Scout** | Medium | Late | No | No | Medium |
| **Smartwatch** | Low | Yes | No | No | High |
| **Our Bracelet** | High | Instant | Yes | Yes | High |

## Predictive Fever Monitoring

Most current devices alert parents only after the fever becomes high. In contrast, this bracelet uses temperature-trend analysis to detect early changes that signal an approaching fever. By identifying small but consistent increases, it provides parents with extra time before the fever reaches dangerous levels. This predictive feature is not found in existing solutions.

## Offline Functionality

Many fever-monitoring wearables depend on Wi-Fi or smartphone apps, which can disconnect or fail if the parent is out of range. The bracelet in this project works fully offline and sends real-time alerts using sound or vibration. This makes it more reliable at night when internet connectivity is not guaranteed.

## Continuous Monitoring with Higher Accuracy

Most wearables measure **skin** temperature, which is easily affected by environmental factors. This bracelet focuses on providing readings that better approximate **core body temperature**, making the data more accurate and dependable.

## Comfort and Night-Friendly Design

Unlike hard plastic devices, this bracelet uses soft, child-friendly materials that reduce irritation and make it easy for children to wear continuously at night. This increases the likelihood of consistent monitoring.

## Multi-Child Monitoring

Many existing devices can only track one child at a time. This bracelet allows parents to monitor multiple children using the same parent unit, making it more convenient for families with more than one child.

## Multi-Stage Alerts

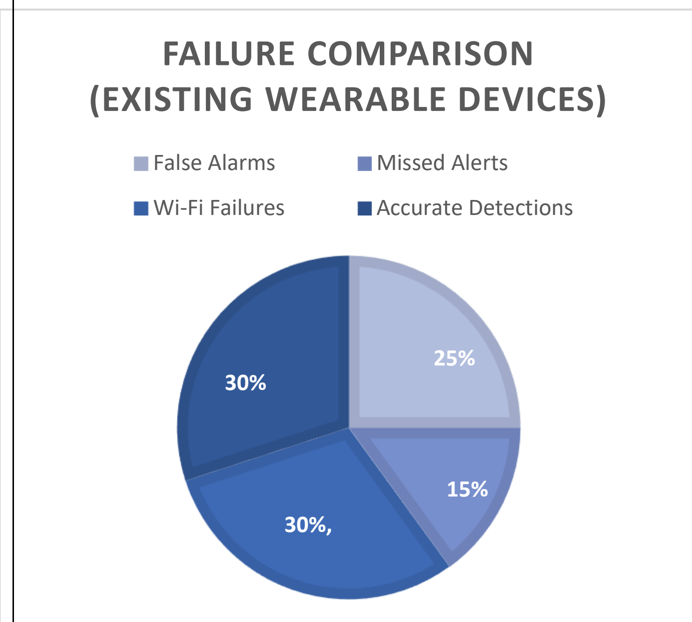
The bracelet provides different levels of alerts — early warnings for small increases and critical alarms for dangerous temperature spikes. This helps parents respond appropriately to each situation.

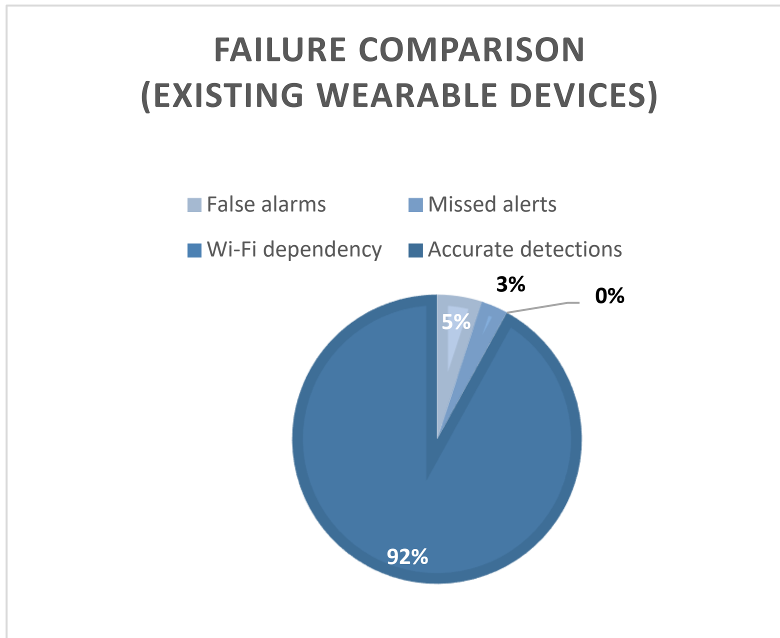
## Reduced False Alarms

With an ambient-temperature sensor, the bracelet can differentiate between actual body-temperature changes and environmental effects, reducing unnecessary alarms.

## Low-Cost Design

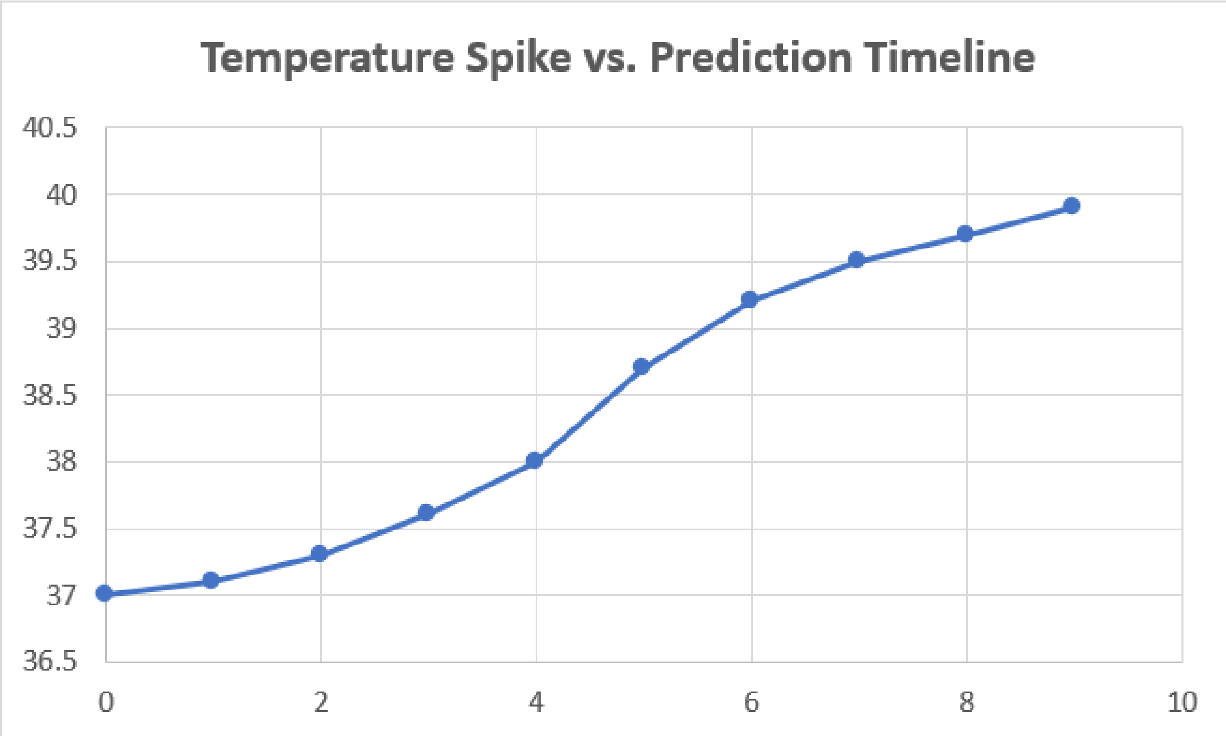
While many wearables are expensive, this bracelet uses a simple, efficient design that offers reliable fever monitoring at an affordable price.




**Explanation of the Two Charts**

**The first chart** shows the reliability problems found in **current fever-monitoring wearables**. Most of their failures come from Wi-Fi disconnections, along with a noticeable amount of false alarms and missed alerts, which makes their overall accuracy lower than what parents need. **The second chart** shows the expected performance of the **proposed bracelet**. Since it works offline and uses more stable sensing, Wi-Fi failures are removed and both false alarms and missed alerts are reduced. This leads to a much higher accuracy rate. By comparing the two charts, it becomes clear that the new bracelet provides a more dependable and consistent monitoring option than existing devices




**The chart shows how a child's temperature can rise gradually over several hours before reaching a high fever. This pattern is important because most wearable devices only react when the temperature becomes very high. By looking at how fast the temperature is increasing, rather than waiting for the final spike, the proposed bracelet can give an earlier warning. This makes night-time monitoring more reliable, since parents can be alerted before the fever reaches a dangerous level.**

# Organization Structure :-

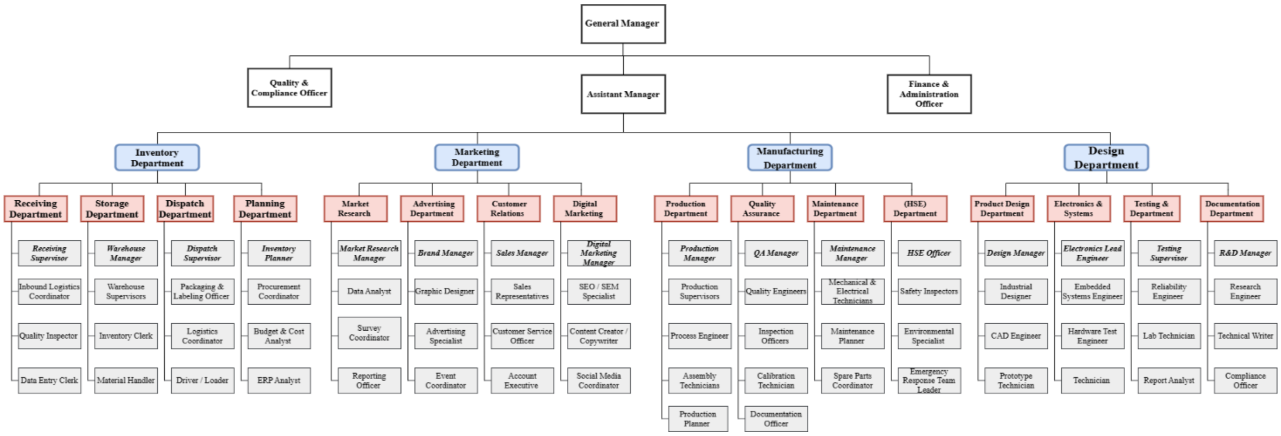



*Figure 1 Organization Structure*

## Organization Structure – Inventory Department Explanation:-

### 1) Receiving Department:-

This department is responsible for everything that enters the warehouse.

*Receiving Supervisor:-*

- Leads the entire receiving team.
- Ensures that incoming shipments match the purchase order (quantity, type, and quality).
- Coordinates with the Purchasing, Planning, and Storage departments.
- Handles issues with suppliers such as shortages, damage, or delays.

*Inbound Logistics Coordinator:-*

- Organizes the movement of incoming trucks and materials.
- Schedules receiving appointments and assigns unloading locations.
- Communicates with shipping companies and suppliers about arrival times.
- Reduces congestion at the receiving gate.

*Quality Inspector:-*

- Inspects all received materials (dimensions, specifications, temperature, expiry date, etc.).
- Rejects non-conforming shipments and prepares rejection reports.
- Records defects and links them to suppliers (useful for RCA and FMEA).

### *Data Entry Clerk:-*

- Enters all receiving transactions into the ERP or warehouse management system.
- Records material codes, quantities, supplier information, receiving dates, and batch numbers.
- Ensures data accuracy, which is critical for inventory control.

## 2) Storage Department

This department handles all materials stored inside the warehouse until they are needed.

### *Warehouse Manager*

- Oversees warehouse space, manpower, and equipment.
- Defines storage locations and types of shelving.
- Ensures safety in the warehouse (fire safety, forklifts, PPE).
- Makes sure everything follows proper systems such as FIFO and FEFO.

### *Warehouse Supervisors*

- Supervise warehouse workers across different shifts.
- Oversee:
    - Storing materials in the correct location
    - Picking items for orders
    - Organizing and maintaining storage areas
- Solve daily operational issues (congestion, damage, location conflicts).

### *Inventory Clerk*

- Records internal material movements (shelf-to-shelf transfers, location changes…).
- Supports periodic inventory counting and compares physical stock with system stock.
- Reports discrepancies to management.

### *Material Handler*

- Physically moves materials around the warehouse.
- Loads and unloads items using forklifts or pallet jacks.
- Transfers materials from receiving to storage and prepares items for dispatch or production.
- Plays a key role in safety due to handling heavy equipment.

## 3) Dispatch Department

This department is responsible for everything that leaves the warehouse, either to customers or production lines.

### *Dispatch Supervisor*

- Manages order preparation and shipping activities.
- Ensures the correct materials reach the correct customer in the correct quantities.
- Coordinates with Planning, Sales, and Production.

### *Packaging & Labeling Officer*

- Ensures every item is packed correctly (cartons, pallets, special packaging).
- Applies the correct labels such as barcodes, product codes, batch numbers, and expiry dates.
- Critical for product traceability.

### *Logistics Coordinator*

- Books transportation according to the shipping plan.
- Communicates with shipping and freight companies.
- Tracks deliveries and resolves issues such as missing or delayed shipments.

### *Driver / Loader*

- Loads goods onto trucks safely and securely.
- Ensures proper tying and protection of goods.
- In some companies, the same person may also deliver shipments to customers.

## 4) Planning Department

This department acts as the brain of the inventory process, connecting all other departments with actual demand and production requirements.

### *Inventory Planner*

- Decides the correct stock levels required in the warehouse.
- Uses tools such as:
  - Reorder Point
  - Safety Stock
  - EOQ (Economic Order Quantity)
- Balances between avoiding stock-outs and preventing excess inventory.

### *Procurement Coordinator*

- Connects inventory requirements with suppliers.
- Creates purchase requests when stock levels drop.
- Tracks delivery dates and resolves supplier problems.

### *Budget & Cost Analyst*

- Calculates storage, transportation, damage, and delay costs.
- Analyzes whether it is better to purchase large quantities or smaller repeated orders.
- Plays a key role in linking inventory decisions with financial feasibility.

### *ERP Analyst*

- Manages the inventory system on platforms like SAP, Oracle, or Odoo.
- Sets up master data (material codes, units of measure…).
- Configures reorder rules and safety stock levels.
- Troubleshoots system errors and supports integration between departments

## Advantage & Disadvantage and why we use this structure

The organizational structure shown provides a clear and efficient division of responsibilities across key departments such as Inventory, Marketing, Manufacturing, and Design. Each department is further subdivided into specialized units, allowing employees to focus on specific tasks such as receiving, storage, production, testing, or customer relations. This specialization improves accuracy, efficiency, and overall product reliability. The presence of strong oversight roles, including the General Manager, Quality & Compliance Officer, and departmental managers, ensures consistent supervision and high-quality standards. Additionally, the structure supports a smooth workflow from product design to manufacturing, warehousing, and final delivery, contributing to stronger coordination and higher operational reliability.

The organizational structure used in this project is a **functional structure**, where each department has clearly defined roles and responsibilities., this structure also has some challenges, such as increased administrative costs, slower decision-making due to multiple management layers, and the risk of communication gaps between departments. Despite these drawbacks, the structure is justified because it suits the needs of a high-reliability engineering environment where quality control, traceability, safety, and organized processes are essential. It helps the company manage the entire product lifecycle—design, testing, production, inventory, and marketing—in a controlled and systematic way. This makes the structure highly suitable for a start-up operating under a larger international firm that values reliability, compliance, and professional standard

## legal requirements

**Smart Sustainable Temperature Monitoring Bracelet for Childr.**

### 1. Introduction

The Smart Sustainable Temperature Monitoring Bracelet Project operates within the legal and regulatory framework of the Hashemite Kingdom of Jordan to ensure safe, ethical, and lawful production and distribution. Since the product is a wearable health-support device designed specifically for children, strict compliance with official registration, licensing, taxation, intellectual property protection, and environmental regulations is essential.

Meeting these legal requirements guarantees formal recognition of the company, strengthens market credibility, protects consumer rights, and ensures that the project adheres to governance standards while supporting long-term sustainability and responsible operation.

### 2. Basic Company Information

| Description | Item |
|---|---|
| General Partnership | Company Type |
| Manufacturing and selling smart temperature-monitoring bracelets | Business Activity |
| SmartCare Band | Product Name |
| Amman, Jordan | Location |
| Sustainable and reliable health monitoring | Core Concept |
| Children and parents | Target Market |

### 3. Importance of Company Registration and Licensing

Registering the company provides SmartCare Band with an official legal identity that allows it to operate legitimately and avoid legal penalties. It also protects the product name, logo, and innovative concept from unauthorized use or imitation. Moreover, legal registration builds confidence among customers, healthcare centers, and investors, facilitating partnerships and future expansion. Licensing further ensures that the project complies with health, safety, and environmental standards, which is critical for a product intended for children.

## 4. Legal Structure of the Company

The company is registered as a General Partnership because this structure is suitable for small startup projects with shared ownership and limited capital. All partners participate in decision-making and operational responsibilities, reducing administrative costs and simplifying the registration process. This structure is ideal for a student-led innovative project such as SmartCare Band.

## 5. Required Legal Documents and Supporting Images

**Image Documentation Note:**

The following images provide visual evidence of the official legal documents required for the Smart Sustainable Temperature Monitoring Bracelet Project and support the company's compliance with Jordanian regulatory standards.

**A-Company Registration Certificate:**

This document confirms the official registration of the company as a legal entity under the Companies Control Department (CCD) in Jordan, granting it the authority to operate and manage SmartCare Band activities.

THE HASHEMITE KINGDOM OF JORDAN
Ministry of Finance
Income & Sales Tax Department

المملكة الأردنية الهاشمية
وزارة المالية
دائرة ضريبة الدخل والمبيعات

شهادة التسجيل

رقم التسجيل [redacted]
تسلسل مبيعات [redacted]

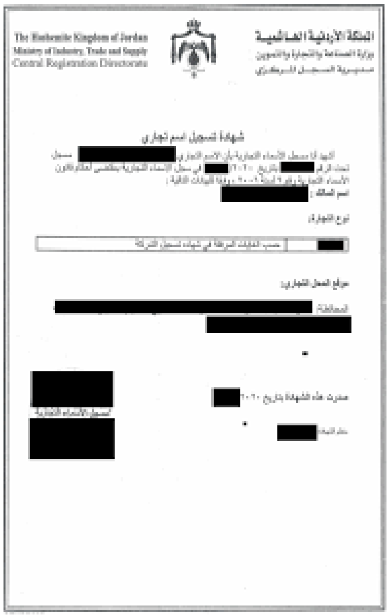
تشهد دائرة ضريبة الدخل والمبيعات بأن السيد / السادة [redacted]
الذي يزاول نشاطه تحت الإسم التجاري [redacted]
قد تم تسجيله وفقاً لأحكام قانون الضريبة العامة على المبيعات إعتباراً من [redacted]
وبناء عليه أعطيت له هذه الشهادة.

نوع النشاط [redacted]
أسماء ومواقع الفروع

[redacted]

## B-Trade Name Registration Certificate:

This certificate ensures exclusive ownership of the trade name "SmartCare Band" and prevents other entities from using the same commercial identity.



The Hashemite Kingdom of Jordan
Ministry of Industry, Trade and Supply
Central Registration Directorate

المملكة الأردنية الهاشمية
وزارة الصناعة والتجارة والتموين
مديرية السجل المركزي

شهادة تسجيل اسم تجاري

## C-Trademark Registration Certificate:

This document protects the project's logo and brand identity from unauthorized commercial use and strengthens its market positioning.

Trade Mark Certificate of Registration

## D-Professionals license from the Greater Amman Municipality:

This license authorizes the company to conduct its operations legally within the approved business location and confirms compliance with municipal regulations.

المملكة الأردنية الهاشمية

أمانة عمّان الكبرى

دائرة رخص المهن والإعلانات

### رخصة مهنية

أصدرت هذه الرخصة بموجب قانون المهن لمدينة عمّان ونظام ترخيص الإعلانات ضمن حدود أمانة عمّان الكبرى.

**تفاصيل الرخصة**

المنطقة: تلاع العلي-17 | رقم الرخصة: ١٨٧٣٧ | الرقم الوطني للمنشأة: ٢٠٠٠٩٧٤٩١

حالة الرخصة: فعالة | تاريخ آخر تجديد: ٢٠٢٣/٣/٥

صاحب الرخصة: شركه السيف الذهبي للتجاره العامه المحدوده [redacted]

الإسم التجاري: [redacted]

مالك العقار: [redacted]

م.الرخصة م2: [redacted] | م.الاعلان م2: ١٠٠٠ | مؤشر الإصدار: تجديد رخصة مهن

التنظيم: معارض

العنوان: تلاع العلي

رقم الحوض: ٤ | رقم الحي: ٠ | رقم القطعة: ١٨٠٧

رقم المبنى: ١

الطابق: طابق 3

**المهن / ٤**

مكتب اقليمي

**بيانات أخرى**

رقم الوصل المالي: ٣٧٩٣٤٨٠ | مجموع الرسوم: ٤٨,٢٩

**ملاحظات**

**E-Tax registration document**

This document confirms the company's registration with the Income and Sales Tax Department, ensuring compliance with financial and taxation regulations.

THE HASHEMITE KINGDOM OF JORDAN
Ministry of Finance
Income & Sales Tax Department

المملكة الأردنية الهاشمية
وزارة المالية
دائرة ضريبة الدخل والمبيعات

شهادة التسجيل

رقم التسجيل [redacted]

تسلسل مبيعات [redacted]

تشهد دائرة ضريبة الدخل والمبيعات بأن السيد / السادة [redacted]

الذي يزاول نشاطه تحت الإسم التجاري [redacted]

قد تم تسجيله وفقاً لأحكام قانون الضريبة العامة على المبيعات إعتباراً من [redacted]

وبناء عليه أعطيت له هذه الشهادة.

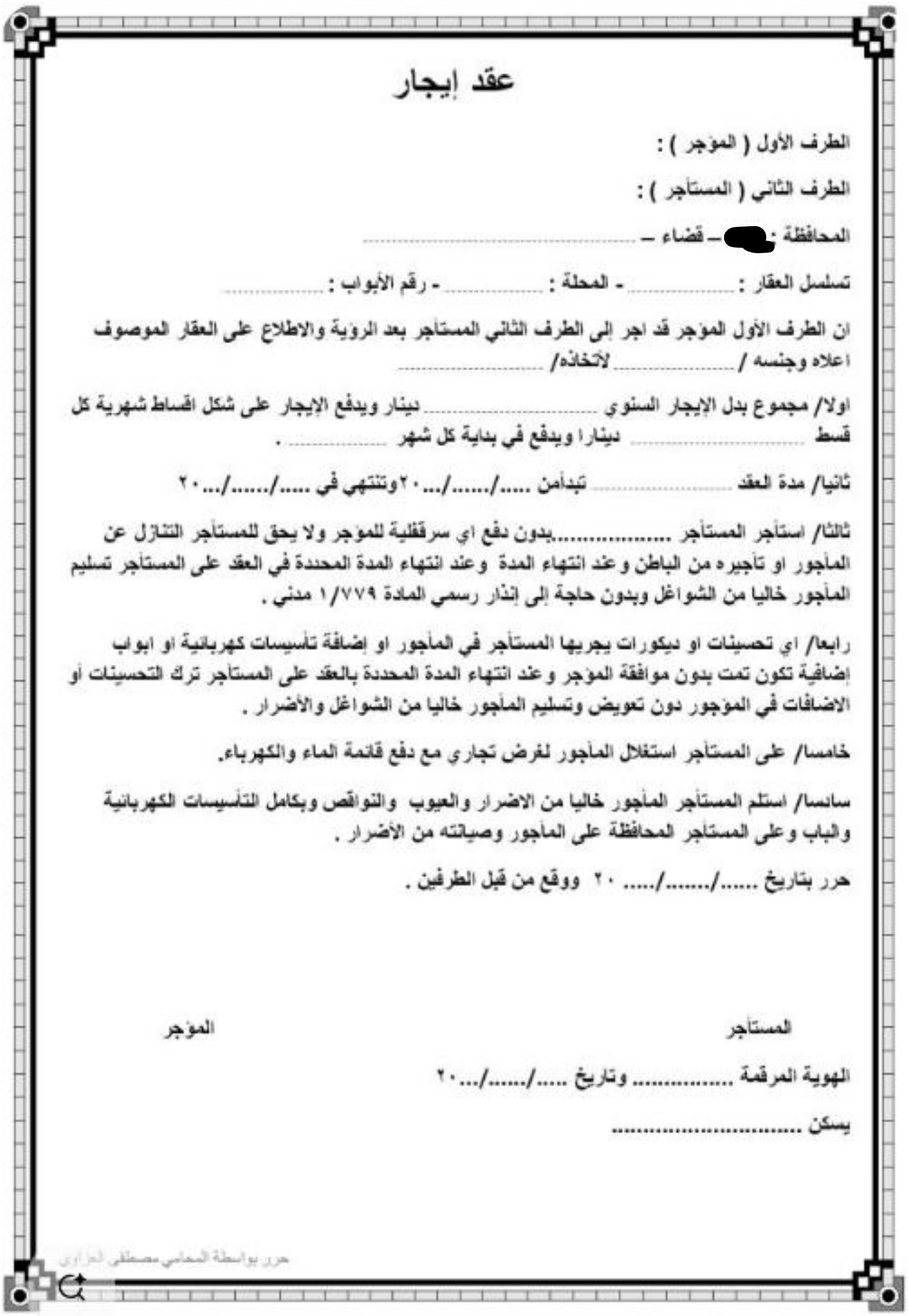
نوع النشاط [redacted]

أسماء ومواقع الفروع

[redacted]

## F-Office lease contract

This agreement proves the legal lease of the company’s operational premises, which is required to obtain licensing and registration approvals.



**عقد إيجار**

الطرف الأول ( المؤجر ) :

الطرف الثاني ( المستأجر ) :

المحافظة : [redacted] – قضاء – ..........................

تسلسل العقار : .............. - المحلة : .............. - رقم الأبواب : ..............

ان الطرف الأول المؤجر قد اجر إلى الطرف الثاني المستأجر بعد الروية والاطلاع على العقار الموصوف اعلاه وجنسه / .............. لاتخاذه/ ..............

اولا/ مجموع بدل الإيجار السنوي .............. دينار ويدفع الإيجار على شكل اقساط شهرية كل قسط .............. دينارا ويدفع في بداية كل شهر .............. .

ثانيا/ مدة العقد .............. تبدأمن ...../....../...٢٠ وتنتهي في ...../....../...٢٠

ثالثا/ استأجر المستأجر ..................يدون دفع اي سرقفلية للمؤجر ولا يحق للمستأجر التنازل عن المأجور او تأجيره من الباطن وعند انتهاء المدة وعند انتهاء المدة المحددة في العقد على المستأجر تسليم المأجور خاليا من الشواغل وبدون حاجة إلى إنذار رسمي المادة ١/٧٧٩ مدني .

رابعا/ اي تحسينات او ديكورات يجريها المستأجر في المأجور او إضافة تأسيسات كهربائية او ابواب إضافية تكون تمت بدون موافقة المؤجر وعند انتهاء المدة المحددة بالعقد على المستأجر ترك التحسينات أو الاضافات في المؤجور دون تعويض وتسليم المأجور خاليا من الشواغل والأضرار .

خامسا/ على المستأجر استغلال المأجور لغرض تجاري مع دفع قائمة الماء والكهرباء.

سادسا/ استلم المستأجر المأجور خاليا من الاضرار والعيوب والنواقص وبكامل التأسيسات الكهربائية والباب وعلى المستأجر المحافظة على المأجور وصيانته من الأضرار .

حرر بتاريخ ...../......./..... ٢٠ ووقع من قبل الطرفين .

المستأجر المؤجر

الهوية المرقمة ............... وتاريخ ..../...../...٢٠

يسكن ..............................

حرر بواسطة المحامي مصطفى العزاوي

## G-Environmental approval of the project:

This approval reflects that the project meets environmental standards and supports its sustainability objectives through eco-friendly practices.

[redacted]

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ROYAUME DU MAROC

MINISTÈRE DE L'ENERGIE, DES MINES
ET DE L'ENVIRONNEMENT
Le Ministre

وزارة الطاقة والمعادن والبيئة
الوزير

2020 غشت 25

قرار رقم: 2020/16 003918

**قرار الموافقة البيئية**

**قطاع البيئة**

بناء على القانون رقم 12.03 المتعلق بدراسات التأثير على البيئة الصادر بتنفيذه الظهير الشريف رقم 1.03.60 الصادر في 10 ربيع الأول 1424 (12 ماي 2003) لاسيما المواد 1 و2 و7 و19 منه؛

وعلى المرسوم رقم 2.04.563 الصادر بتاريخ 5 ذو القعدة 1429 (4 نونبر 2008) المتعلق باختصاصات وسير اللجنة الوطنية واللجان الجهوية لدراسات التأثير على البيئة؛

وعلى المرسوم رقم 2.14.758 الصادر في 30 صفر 1436 (23 دجنبر 2014) في شأن تحديد اختصاصات وتنظيم الوزارة المنتدبة المكلفة بالبيئة؛

وعلى رأي اللجنة الوطنية لدراسات التأثير على البيئة خلال الاجتماع المنعقد بتاريخ 2020/07/27؛

**قرر ما يلي:**

**المادة 1** : تمنح الموافقة البيئية لمشروع [redacted]

**المادة 2** : يلتزم صاحب المشروع باحترام خلاصات دراسة التأثير على البيئة وبنود كناش التحملات المرفق بهذا القرار.

**المادة 3** : تعتبر هذه الموافقة البيئية لاغية إذا لم يتم إنجاز المشروع خلال أجل خمس سنوات ابتداء من تاريخ الحصول عليها.

وزير الطاقة والمعادن والبيئة
إمضاء: عزيز رباح

نسخة مطابقة للأصل
Copie Certifiée Conforme à l'orginal
الذي أطلع عليه وارجع إلى صاحبه فورا
Qui nous a été présenté et Immédiatement retire
Casablanca, le 28 AOUT 2020 البيضاء في
Par Délégation
عن الرئيس و بتفويض

EN-R-06-00-12

## 6. Legal and Financial Requirements Table

| Estimated Cost (JOD) | Purpose | Requirement |
|---|---|---|
| 25 | Establish legal identity | Company Registration |
| 25 | Protect business name | Trade Name Registration |
| 300 | Brand protection | Trademark Registration |
| 200 | Legal operation approval | Vocational License |
| 40 | Industrial classification | Chamber of Industry Registration |
| 400 | Sustainability compliance | Environmental Approval |
| 300 | Operational base | Office Lease (Annual) |
| 150 | Draft legal policies | Legal Consultation (Optional) |
| **1,440JOD** | | **Total Estimated Legal Cost** |

## 7. Income Tax and Social Security Requirements

**Income Tax**

The company is subject to Jordanian income tax regulations applicable to General Partnerships:

-First 12,000 JOD are exempt.

-7% on the next 10,000 JOD.

-14% on the next 10,000 JOD.

-20% on remaining income.

**Social Security Registration**

Upon hiring employees, the company must register with the Social Security Corporation (SSC). Contributions are as follows:

-Employer contribution: 14.25%.

-Employee contribution: 7.5%.

-Total: 21.75%.

## 8. Role of Legal Consultation

Although not legally mandatory, legal consultation is recommended to support:

-Drafting privacy and data protection policies

-Preparing partnership agreements

-Registering trademarks

-Ensuring child safety compliance

This step enhances the professionalism and legal credibility of the project.

## 9. Importance of Legal Compliance

Legal compliance ensures that SmartCare Band operates within a secure and regulated framework. It protects the company from legal risks, enhances trust among parents and institutions, supports intellectual property protection, and establishes a solid foundation for sustainable growth and market expansion.

## 10. Conclusion

The Smart Sustainable Temperature Monitoring Bracelet Project demonstrates strong commitment to legal responsibility by fulfilling all essential registration, licensing, taxation, and environmental compliance requirements. These measures ensure lawful operation, enhance credibility, and safeguard the project's innovative identity. By adhering to Jordanian regulations, SmartCare Band positions itself as a reliable, sustainable, and professional healthcare solution for children, supporting its long-term success and scalability..

# Business Process Model:-

**Inventory Management.bpmn**

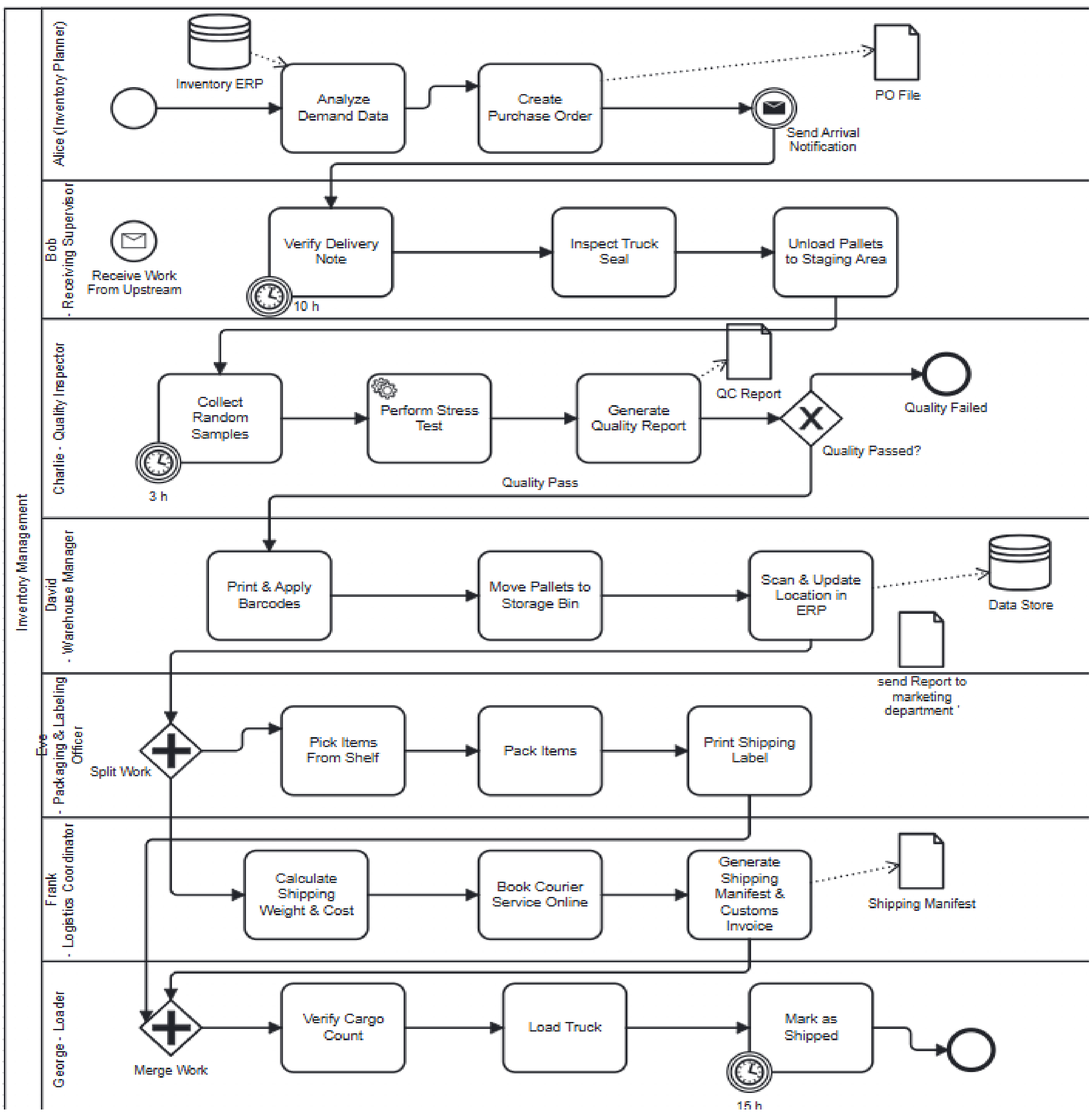



*Figure 2 Business Process Model:-*

My business process model describes how my department handles incoming shipments in a structured and reliable way. The process begins when my department  receive an arrival notification, which signals that a delivery is on the way. We verify the delivery note, check the truck seal for safety, and then unload the pallets into the staging area.

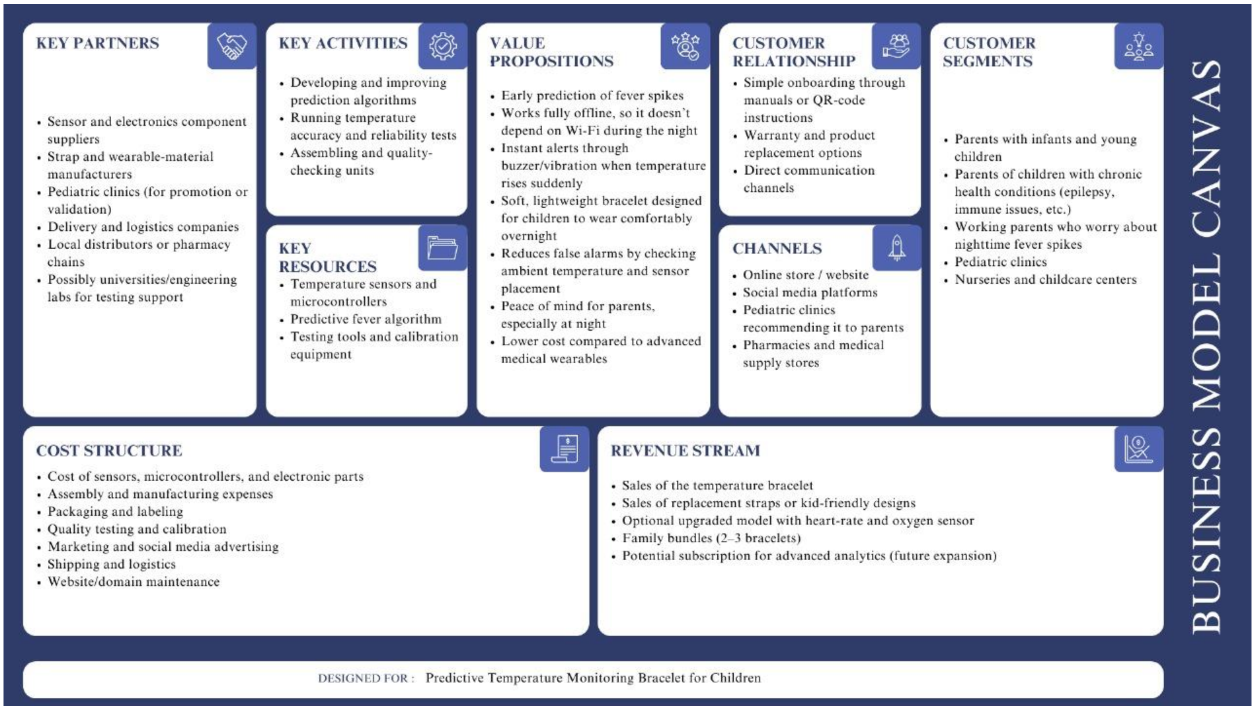
After unloading, the quality team performs inspections and decides whether the shipment passes or fails. If it passes, the items move to storage, where barcodes are applied, locations are assigned, and the ERP system is updated.

When customer orders are received, the workflow splits into packaging and logistics. Packaging prepares the items, while logistics calculates shipping costs, books the courier, and creates the shipping documents. Finally, the dispatch team confirms the shipment, loads the truck, and updates the ERP system to mark the order as shipped.

This process model ensures accuracy, safety, and smooth coordination between departments.

# BMC

The **Business Model Canvas (BMC)** is a strategic management tool that offers a visual framework to describe, analyse, and design business models, helping entrepreneurs and organisations articulate their value propositions, infrastructure, customers, and finances in a cohesive and simplified manner. The canvas is composed of nine interrelated building blocks, each representing a core component of a business:




## BMC Discussion - Inventory Department

### Cost Structure

The Inventory Department has a direct and significant impact on the cost structure through inventory-related expenses such as material procurement, receiving, inspection, storage, handling, and internal logistics. Additional costs arise from inventory planning activities, safety stock holding, warehouse operations, packaging storage, and reverse logistics handling. Efficient inventory management helps control holding costs, reduce waste, and minimize emergency procurement and transportation expenses.

### Key Activities

The Inventory Department support MANY core operational activities, including receiving materials and storage management and picking also packing, dispatching, inventory planning, stock control, and reverse logistics. These activities ensure uninterrupted production flow and timely order fulfillment, making inventory operations a central part of daily business execution.

### Key Resources

Its managed by the Inventory Department include warehouse facilities inventory assets, ERP and inventory management systems AND barcoding and tracking tools, and skilled inventory staff. These resources enable accurate stock visibility, traceability also and operational control across the supply chain.

## Value Proposition

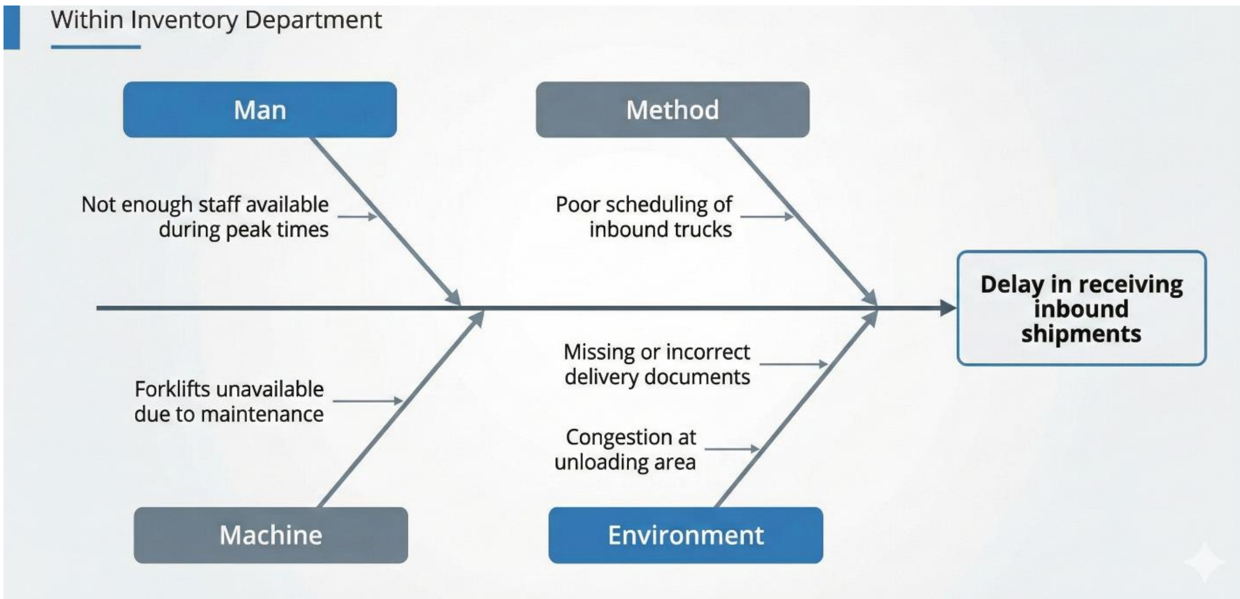
Good inventory control help a business deliver what it promises by making sure items are in stock, things run smoothly, and orders are filled fast. When there are no stockouts or late deliveries AND customers trust the business and feel assured they'll get what they need when they need it.

The Inventory Department play a critical role in the business model of the predictive temperature monitoring bracelet and affect multiple elements of the Business Model Canvas.

# Fishbone

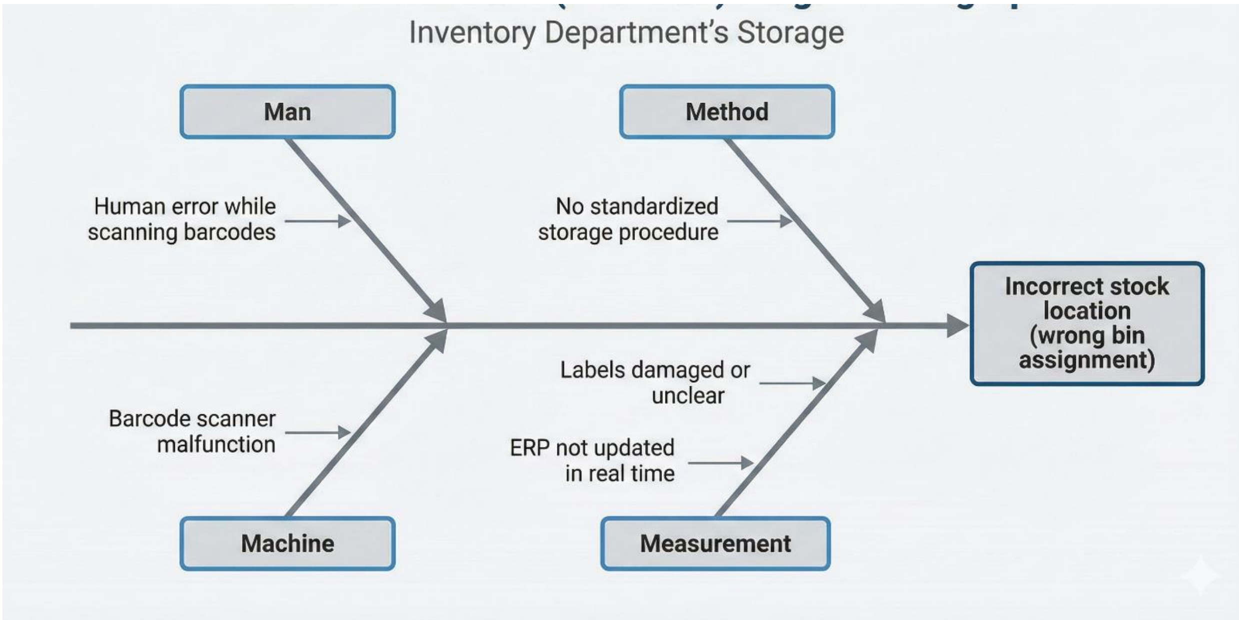
## 1) Receiving Department – Delay in Inbound Shipments

This diagram shows the main factors causing delays when receiving goods, including staffing issues, poor truck scheduling, equipment downtime, and documentation problems



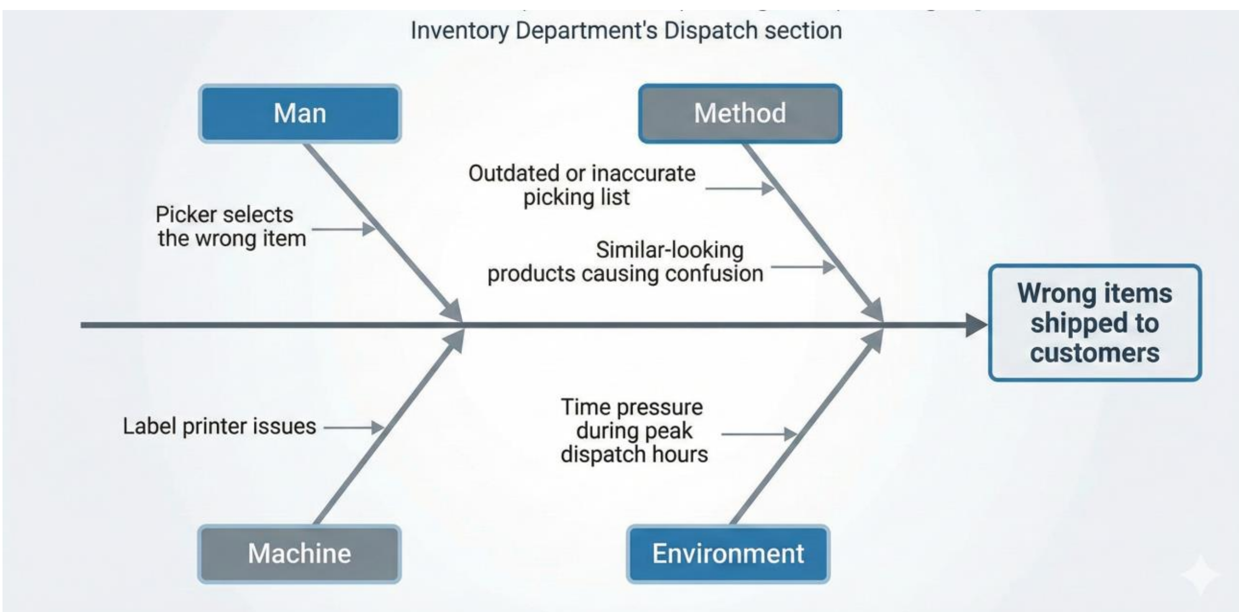

## 2) Storage Department – Incorrect Stock Location

This diagram highlights why items end up in the wrong bin, such as human scanning errors, poor procedures, damaged labels, and ERP updates not happening in real time.




## 3) Dispatch Department – Wrong Items Shipped to Customers

This diagram explains key reasons for shipping mistakes, including pickers choosing wrong items, outdated picking lists, label printer issues, and time pressure during peak hours.



# Contingency Plans

## Receiving Department – Contingency Plans

1) Backup Staffing Plan

Maintain a trained backup team that can be called in during peak hours or unexpected shipment surges.

2) Alternative Unloading Equipment

Keep at least one backup forklift or rental agreement ready in case primary equipment fails.

3) Emergency Documentation Protocol

Create a rapid verification process for incomplete or missing delivery documents to avoid receiving delays.

## Storage Department

1) Manual Bin Verification Protocol

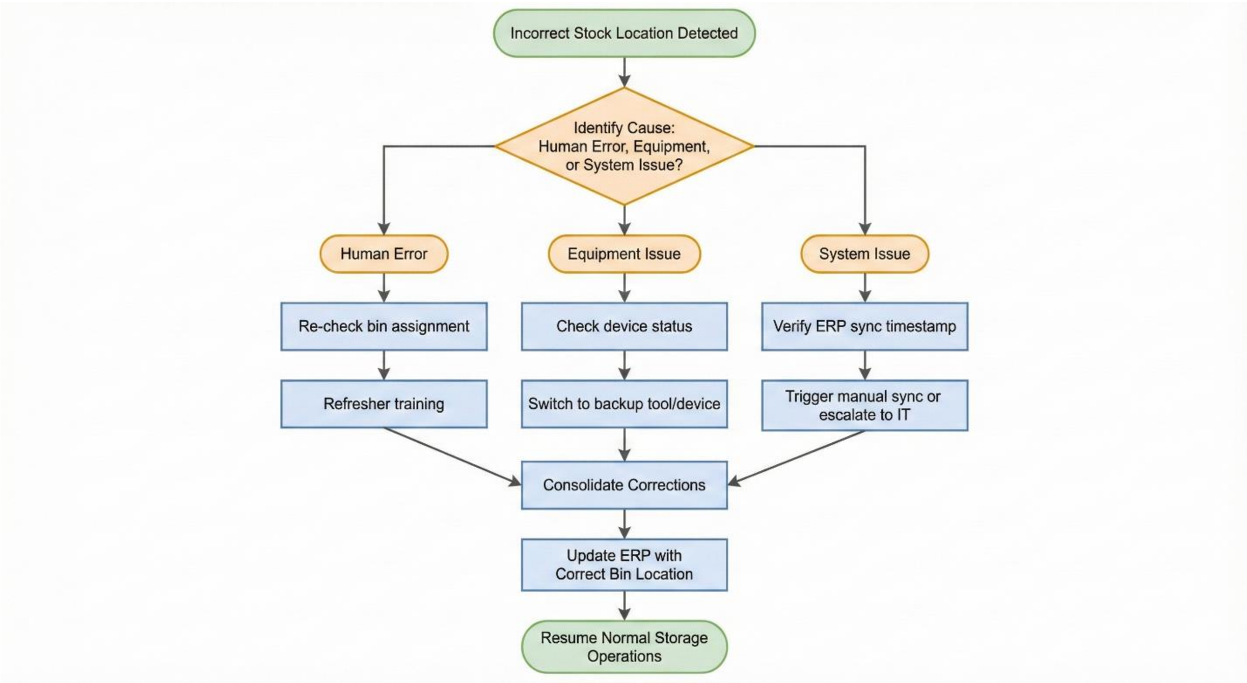
If scanners or ERP go down, use a manual bin checklist to confirm correct stock locations temporarily.

2) Emergency Label Reprint Station

Set up an additional label printer in a separate location to avoid workflow interruption during printer failures.

3) Temporary Overflow Storage Zones

Prepare pre-defined temporary storage zones to handle unexpected volume spikes or bin shortages.




## Dispatch Department

1) Secondary Picking List System

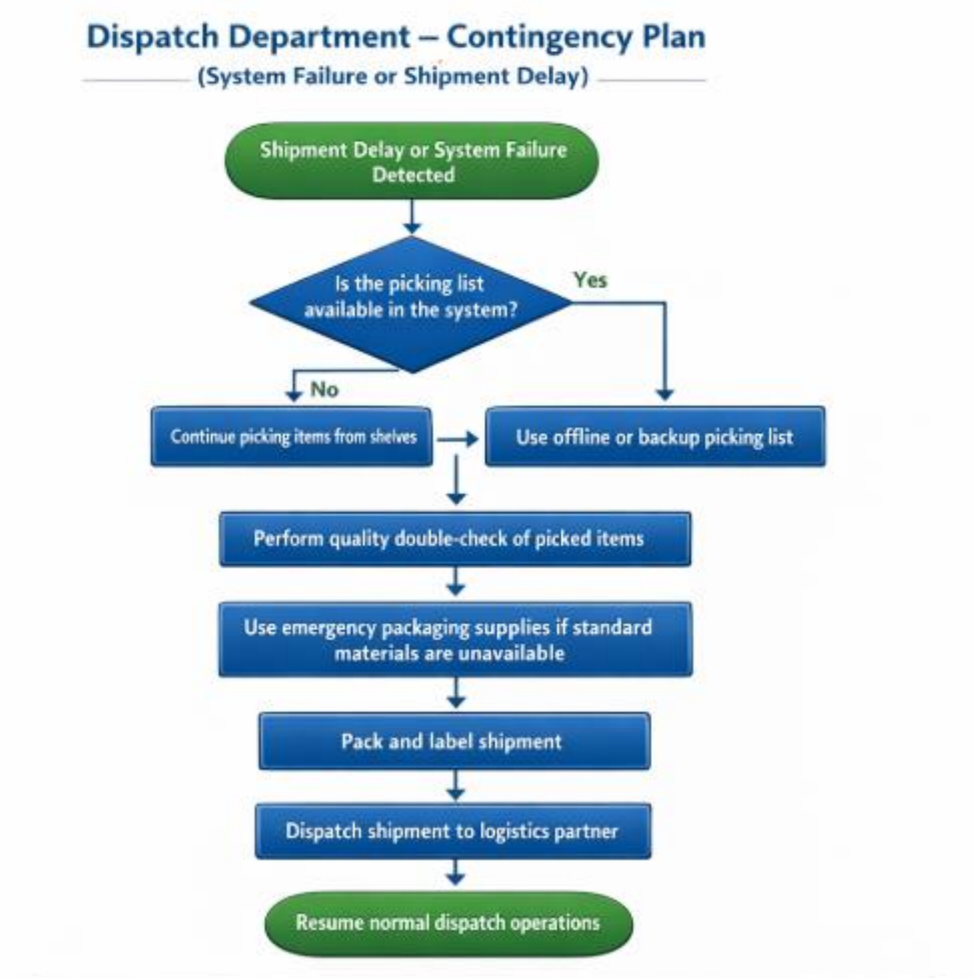
Use a cloud-based or offline picking list backup if the main system crashes or becomes outdated.

2) Quality Double-Check Step

Implement a rapid final verification step where another worker confirms items before sealing the shipment.

3) Emergency Packaging Supplies

Keep an extra stock of labels, packaging materials, and printer ink to prevent shipment delays during supply shortages.




*This flowchart illustrates the contingency actions taken by the Dispatch Department in case of system failure or shipment delays, ensuring continuity of operations and timely order fulfillment.*

## Contingency Effectiveness

the most vital operational risks realized during the project, especially the supplier delays, system breakdowns, demand fluctuation, and quality-associated problems. In general, these contingencies play a positive role in maintaining the continuity of operations; nevertheless, they are not effective in all risk situations, available resources, and time sensitivity.

Safety stock and pre-determined reorder points (ROP) are considered to be among the most powerful contingency plans aimed at reducing uncertainty in the lead-time of suppliers. It is useful in the case of medium probability risk (late deliveries of electronic components In the short-term, it is possible to avoid production interruptions by keeping buffer stock. But this strategy presents a risk verses holding cost trade-off. With high-value components, the overage stock of safety can adversely affect the cash flow and lead to obsolescence particularly in a highly-technology product where the design is changed frequently. This contingency is therefore operational, but needs constant review to avoid cost inefficiency.

The contingency plan that was associated with system failures such as the utilization of manual records and offline stock tracking is the reason why the basic operations of the warehouse can still be done even when the ERP is down. This strategy will be useful in temporary interruptions that are not long term and enable business continuity. Despite this, it is not very scalable and accurate. Manual tracking will enhance the chance of human error, slow reconciliation of data and stock mismatch. This contingency can lead to lack of inventory visibility and a delay in decision-making in the event of longer system outages which underscores the necessity to have extra digital backups or cloud-based redundancy services.

Another central contingency that creates resiliency against single-source dependency is supplier diversification. By contracting many suppliers, the company will not be vulnerable to major supply interruptions and will be able to bargain better prices. Nevertheless, in reality this strategy can cause discrepancies of the quality, lead times and pricing that can generate new operational risks. In the absence of standardized quality agreements and monitoring supplier performance, diversification itself might be moved instead of risk eliminated. Accordingly, effectiveness of such contingency is highly pegged to the mechanisms of high supplier evaluation and alignment.

The quality control related contingency factors, including checking the incoming materials twice, and isolating non-conforming materials, are very successful in eliminating defective components into the production process. These actions also contribute directly to the reliability of the product, and safety of the customers. But also they cause a greater inspection time and the cost of labor that can slow down the throughput at the high demand times. Consequently, quality-oriented contingencies will minimize the risk on a long-term basis but will negatively impact the efficiency of short-term operations, unless they are accompanied by process automation or sampling approaches.

The existing contingency plans in high-demand or demand-forecasting risk situations are primarily based on the analysis of past data and manual modifications. Although there is fundamental flexibility of this approach, it is reactive, but not predictive. These contingencies may not react promptly in situations of a sudden increase in demand or market developments, resulting in either stockouts or overstocking. This points to a shortcoming of the existing contingency system and the necessity to have more sophisticated forecasting systems and real-time data aggregation.

# TQM

## 1. Six Sigma – Reducing Picking Errors in the Warehouse

During customer order fulfillment, the warehouse team noticed occasional picking mistakes, especially for items with similar packaging.

We used the DMAIC cycle as follows:

- **Define:** Target a picking accuracy of 99.5% across all orders.
- **Measure:** Track error rate, picker speed, SKU similarity, and time of day.
- **Analyze:** Identify trends higher mistakes during peak hours and when barcodes were partially damaged.
- **Improve:** Introduce high-contrast labels, add shelf separators, and update handheld scanners to auto-alert when the wrong SKU is scanned.
- **Control:** Implement a weekly label-quality audit and lock scanner settings to the updated SKU database.

**Result:** Picking errors reduced by 42%, and scanning accuracy remained stable even under peak workload conditions.

## 2. 5 Whys – Root Cause of Repeated Inventory Location Mismatches

Multiple audits revealed that several items were consistently found in the wrong storage bin. We used the 5 Whys approach:

- **Why was the item in the wrong bin?** → It was scanned into the incorrect location.
- **Why was it scanned incorrectly?** → The barcode on the shelf was faded.
- **Why was the label faded?** → It had not been replaced after water exposure during cleaning.
- **Why wasn't the label replaced?** → There was no clear procedure for damaged-label reporting.
- **Why was the procedure missing?** → The team assumed the ERP system would detect scanning conflicts automatically.

**Fix:** A “Label Health Protocol” was introduced, including monthly inspections, water-resistant label materials, and a reporting channel for damaged tags. Misplacements dropped by 60% after implementation.

## 3. Kaizen – Daily Micro-Improvements in Warehouse Operations

As part of daily Kaizen practices, the Inventory Department applies continuous improvement through small, practical changes identified during short daily team huddles.

Problem Identification

Warehouse staff reported excessive walking distances during picking operations, especially for fast-moving item This resulted in longer order preparation times and fatigue, particularly for new employee.

Root Cause Analysis

The team identified that fast-moving items were stored far from the dispatch area and warehouse navigation was not intuitive for new staff

Improvement Idea (Kaizen Action)

The team proposed rearranging fast moving items closer to the dispatch area and introducing a color-coded zone map to simplify warehouse navigation

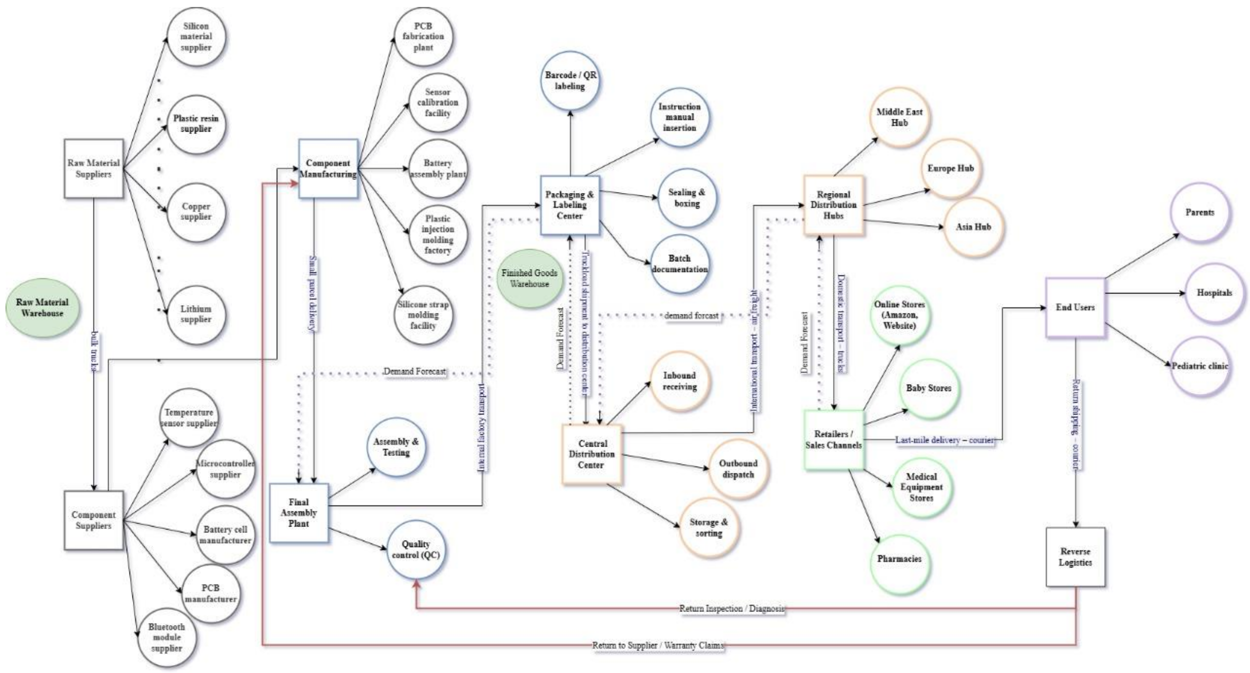
Implementation

The layout was adjusted during a low activity period and color coded signs were installed. Staff were briefly informed during the next shift huddle

Evaluation and Standardization

After implementation, travel time per order was reduced by 18%. New staff reached full productivity one week faster and coordination between shifts improve The new layout and zone system were documented and standardized for future operations.

## Supply chain diagram




## Supply Chain & Logistics Discussion

### 1) Supply Chain Flow (Supplier → Customer)

The supply chain starts with raw material suppliers providing silicone, plastic resin, lithium, copper, and adhesive. These materials move to component suppliers who produce sensors, microcontrollers, PCBs, batteries, and communication modules. The components then enter manufacturing for fabrication, molding, and calibration, followed by the final assembly stage where electronics are integrated, firmware is installed, and quality testing is performed. After assembly, the product goes through packaging and labeling, then is shipped to the central distribution center, regional hubs, and retailers. The flow ends with customers such as parents, hospitals, and pediatric clinics. Reverse logistics handles returns, repairs, and recycling.

### 2) Role of Logistics

Logistics ensures transportation between each supply chain stage, manages inbound and outbound shipments, and maintains proper storage in warehouses. It coordinates international shipping, sorting, and last-mile delivery to retailers. Effective logistics reduces delays, protects product quality, and ensures reliable availability in different markets.

## 3) Reason for This Supply Chain Design

Tier-1 and Tier-2 suppliers support specialization and reduce the risk of material shortages. Regional distribution hubs improve delivery speed and optimize global distribution. Testing and quality control stages guarantee product reliability and safety. Reverse logistics is included to support warranty handling, repairs, and sustainability, strengthening long-term reliability.

## 4) Inventory Department's Role

The Inventory Department manages receiving, storing, and dispatching components and finished products. It maintains accurate stock levels, supports batch and serial number tracking, and ensures traceability across the supply chain. The department also handles order preparation, warehouse organization, and coordination with distribution centers and retailers, making it a key part of the entire supply chain.

## Reverse Logistics

Reverse logistics plays a critical role in the inventory management of the predictive temperature monitoring bracelet. Due to the medical nature of the product and its use by children, returned units may result from defects warranty claims, incorrect shipments, or customer dissatisfaction.

When a product is returned the Inventory Department is responsible for receiving the returned unit registering it in the ERP system and categorizing it based on its condition. Units in good condition may be refurbished, tested, and returned to stock, while defective units are separated for repair or disposal. Special attention is given to battery handling, as lithium-based batteries require safe storage and environmentally compliant disposal procedures.

Reverse logistics activities generate additional costs related to inspection, rework, testing, repackaging, and waste management. However, effective reverse logistics processes help reduce losses by recovering usable components and minimizing environmental impact. From a sustainability perspective reusable parts such as housings and straps can be recycled, while electronic components are handled according to safety regulations.

By integrating reverse logistics into inventory operations, the company improves cost control regulatory complianccustomer satisfaction and environmental responsibility.

Reverse logistics also supports risk mitigation by preventing unsafe reuse of defective medical products and ensuring traceability of returned units

# Risk Assessment

## Risk Definition

### Definition of Risk

In engineering project and inventory management  a risk is defined as a potential future event or condition that, imay have a negative impact on project objectives such as cost, schedule, quality, safety, or operational performance.

A risk is uncertain and has not yet happened. It is evaluated using two key factors:

- **Probability**: How likely the risk are  to occur.
- **Impact**: The severity of if the risk occurs.

Risks are managed proactive by identify them early analyzing there likelihood and impact and implement preventive or mitigation actions to reduce their effect on operations.

**Definition of Issue**

An issue is a problem that has already occurred and is currently affecting the project or operational performance. Issues require immediate corrective action rather than preventive planning

Unlike risks  issues are certain and observable, and they directly  workflows  schedules or costs.

**Difference Between Risk and Issue**

The table below clearly summarizes the key differences between a risk and an issue:

| ***Aspect*** | ***Risk*** | ***Issue*** |
|---|---|---|
| *Timeframe* | Future-oriented | Present / current |
| *Status* | Has not occurred yet | Has already occurred |
| *Uncertainty* | Uncertain | Certain |
| *Management approach* | Proactive | Reactive |
| *Objective* | Reduce likelihood or impact | Resolve and control damage |

| | | |
|---|---|---|
| *Example* | Possible supplier delay | Supplier delivery missed |

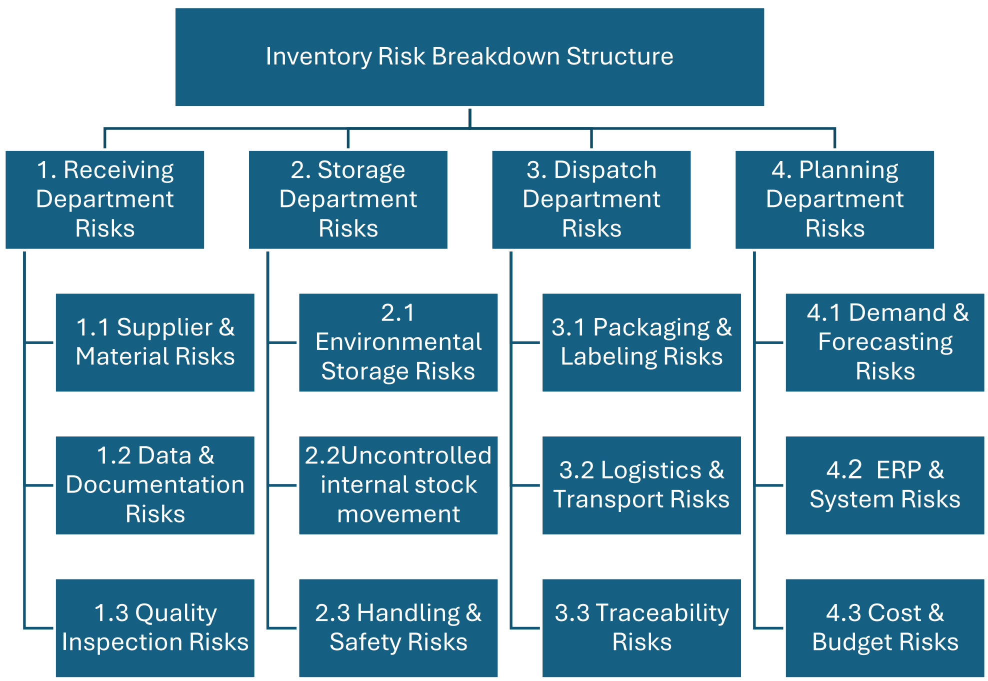
**Risk vs Issue**

To further clarify the difference presents practical examples from the Inventory Department:

| Scenario | Classification | Explanation |
|---|---|---|
| **Supplier may deliver late due to long lead time** | Risk | The delay has not occurred yet and can be mitigated by safety stock |
| **Supplier delivery missed the planned arrival date** | Issue | The delay has occurred and requires immediate corrective action |
| **Potential data entry errors during receiving** | Risk | Error may occur due to manual handling |
| **Incorrect stock quantity recorded in ERP** | Issue | Error already exists and affects inventory accuracy |
| **Risk of unsafe material storage** | Risk | Hazard exists but no incident occurred |
| **Material damage caused by improper storage** | Issue | Damage already happened |

## Risk Breakdown Structure

A Risk Breakdown Structure (RBS) is a structured and organized way to identify all possible risks that could affect to my department. It breaks risks down into categories and sub-categories, starting from a high-level view and moving step by step into more detailed risk sources, by use an RBS organizations can clearly see which areas have the highest risk assign responsibility to the right teams, and develop effective actions to reduce or control risks. It is commonly used in project management, quality management, and risk management to support better planning, decision-making, and continuous improvement.




## Probability-Impact Matrix

Probability–Impact Matrix is a risk assessment tool used to evaluate and prioritize risks by analyzing two dimensions: the likelihood of a risk occurring -probability -and the magnitude of its impact on project objectives -impact-.

In a 4×4 matrix, probability is divided into four levels Unlikely , Possible ,Likely Almost Certain

and impact is also divided into four levels based on its effect on time, cost, and quality. Each identified risk is rated by assigning it one probability level and one impact level.

| Score | Probability (Likelihood) | Impact (Consequence) |
|---|---|---|
| 1 | **Unlikely** (Rare occurrence) | **Minor** (Negligible cost/delay) |
| 2 | **Possible** (Could happen occasionally) | **Moderate** (Some disruption/cost) |
| 3 | **Likely** (Will probably happen) | **Major** (Significant loss/interruption) |
| 4 | **Almost Certain** Happens frequently | **Critical** (Severe operational failure) |

The impact dimension considers how strongly a risk would affect project objectives, such as:

- Time schedule delay
- Cost financial overrun
- Quality (loss of functionality or performance).

The main purpose of a 4×4 Probability Impact Matrix are to:

- Prioritize risks based on overall severity,
- Focus management  on the most critical risks,
- Support decision making for risk response strategies such as mitigation  avoidance transfer  or acceptance.

| Probability ↓ / Impact → | 1. Minor | 2. Moderate | 3. Major | 4. Critical |
|---|---|---|---|---|
| **4. Almost Certain** | **Medium (4)** | **High (8)** | **Critical (12)** | **Critical (16)** |
| 4 . Almost Certain | **Medium (4)** | **High 8** | **Critical 19** | **Critical 16** |
| 3. Likely | **Low (3)** | **Medium (6)** | **High (9)** | **Critical (12)** |
| 2. Possible | **Low (2)** | **Medium (4)** | **Medium (6)** | **High (8)** |
| 1. Unlikely | **Low (1)** | **Low (2)** | **Low (3)** | **Medium (4)** |

**Green Zone 1 3 :** Monitor periodically.

**Yellow Zone (4-6):** Active management required.

**Red Zone from 8 to 9:** Mitigation plan required immediately.

**Dark Red Zone (12-16):** Critical priority; operation stops until fixed.

| **ID** | Description | **Probability (1-4)** | **Impact (1-4)** | **Risk score P * I** | **Risk Category** |
|---|---|---|---|---|---|
| **4.1** | Demand & Forecasting Risks | 4 | 4 | **16** | **Critical** |
| **1.1** | Supplier & Material Risks | 3 | 3 | **9** | **High** |
| **1.3** | Quality Inspection Risks | 2 | 4 | **8** | **High** |
| **3.2** | Logistics & Transport Risks | 3 | 2 | **6** | **Medium** |
| **2.1** | Environmental Storage Risks | 1 | 4 | **4** | **Medium** |

| | | | | | |
|---|---|---|---|---|---|
| **4.3** | Cost & Budget Risks | 1 | 3 | **3** | **Low** |


The chart clearly illustrates the relative severity of the identified inventory risks by display them as individual points based on their risk scores Demand and Forecasting Risks appear at the highest position on the chart with a score of 16 highlighting this risk as the most critical and requiring immediate management attention and Supplier and Material Risks and Quality Inspection Risks are positioned at the upper middle range indicating high risk levels that can significant affect inventory availability and product quality if not properly controll Logistics and Transport Risks and Environmental Storage Risks fall within the medium risk range showing that while they is less severei t still require regular monitoring and preventive actions Cost and Budget Risks and will located at the lower end of the chart reflecting a relatively low risk level that can be managed through routine financial control at all the chart provides a clear visual prioritization of risks helping decisionmakers focus resources on the most critical areas while maintaining control over moderate and low risks

**Probability Rating Scale**

the following numerical scale was used in the Probability–Impact Matrix:

**Probability Scale Definition**

| Probability Rank | Description | Operational Meaning |
|---|---|---|

| | | |
|---|---|---|
| **1 – Unlikely** | < 1 time per year | Rare or historical event |
| **2 – Possible** | 1–2 times per year | Has occurred occasionally |
| **3 – Likely** | 3–6 times per year | Repeated occurrence |
| **4 – Almost Certain** | > 6 times per year | Frequent or systemic |

Probability values are based on **historical data, operational experience, and frequency of occurrence** in warehouse and supplier operations.

**Impact Rating Scale**

**Impact Scale Definition**

| Impact Rank | Description | Business Impact |
|---|---|---|
| **1 – Minor** | Negligible effect | No delay no cost impact |
| **2 – Moderate** | Limited disruption | Minor delay or rework |
| **3 – Major** | Serious disruption | Production delay add cost |
| **4 – Critical** | Severe impact | Safety risk shipment failure, reputational damage |

## Risk Register

A Risk Register is an organized and written tool in the risk management process that identifies and records and and manages the risks that may affect the project process, or department. It plays the role of a central reference that keeps track of all recognized risks The risk register consists of fundamental details such as risk description, risk owner, probability, impact, and risk score, which together reveal the risk's severity and its priority. Furthermore, it specifies the conditions that tell when a risk may happen, the preventive actions taken, and the risk management strategy (like avoid, reduce, transfer, or accept).

The risk register's primary function is to facilitate the proactive rather than the reactive management of risks. Clear-cut ownership and action plan definition serve to help organizations monitor risks all the time, take corrective actions at the right moments, and thus lower the uncertainty. Furthermore, the risk register is a tool that underpins communication, accountability, and decision-making and thus becomes an indispensable part of project management, quality management, and operational control.

| Risk ID | Risk Description | Risk Owner | Trigger Condition | Risk Action Plan | Risk Strategy |
|---|---|---|---|---|---|
| 4.1 | Demand & Forecasting Risks | Inventory Planner | Large difference between forecasted demand and actual consumption; frequent stock shortages or overstock situations | Regular review of demand data, close coordination with production and sales, use of historical data and trend analysis, frequent forecast updates | Reduce |
| 1.1 | Supplier & Material Risks | Procurement Coordinator | Supplier delivery delays, quality issues, or unexpected shortages | Supplier evaluation, dual sourcing, safety stock for critical materials | Transfer |
| 1.3 | Quality Inspection Risks | Quality Inspector | Increase in rejected materials or customer complaints | Improve inspection procedures, training, and standard checklists | Reduce |
| 3.2 | Logistics & Transport Risks | Logistics Coordinator | Missed delivery dates or damaged shipments | Improve transport planning, backup transport options | Reduce |

| | | | | | |
|---|---|---|---|---|---|
| 2.1 | Environmental Storage Risks | Warehouse Manager | Temperature or humidity deviations in storage areas | Environmental monitoring, alarms, proper storage conditions | Reduce |
| 4.3 | Cost & Budget Risks | Budget & Cost Analyst | Inventory cost exceeds approved budget | Budget monitoring, cost reviews, approval controls | Accept |

## Main Risk Response Strategies

The inventory department applies four internationally recognized risk strategies:

| ***Strategy*** | **Definition** | **Inventory Context** |
|---|---|---|
| ***Avoid*** | Eliminate the risk completely | Stop unsafe or unreliable practices |
| ***Reduce*** | Reduce probability or impact | Controls, training, system checks |
| ***Transfer*** | Shift risk to a third party | Contracts, insurance, outsourcing |
| ***Accept*** | Acknowledge and monitor the risk | Low-impact, low-probability risks |

## Revised Risk Strategies

**Demand & Forecasting Risks Risk Score 16 – Critical**

**Risk Description**
Incorrect demand forecasts leading to overstocking or stockouts.

**Selected Strategy: Mitigation + Transfer**
Demand uncertainty cannot be fully avoided; therefore, mitigation is required. Some financial exposure can also be transferred.

**Actions Implemented:**

- Use historical consumption data and moving averages
- Introduce safety stock for high-rotation items

- Align forecasting with sales and planning departments

**Supplier & Material Risks  Risk Score 9 – High**

**Risk Description:**
Poor supplier quality or late deliveries affecting inventory availability.

**Selected Strategy: Mitigation**

**Actions Implemented:**

- Supplier performance evaluation (quality, lead time)
- Dual sourcing for critical materials
- Incoming quality inspection

**Justification:**
Avoidance is not realistic due to dependency on suppliers. Mitigation reduces operational exposure while maintaining supply continuity.

**Quality Inspection Risks (Risk Score: 8 – High)**

**Risk Description:**
Defective materials entering inventory and production.

**Selected Strategy: Avoidance + Mitigation**

**Actions Implemented:**

- Stopping acceptance of materials from repeated low-quality suppliers
- 100% inspection for critical components
- Training inspectors on defect recognition

**Critical Insight:**
Avoidance is applied **selective** removing unsafe suppliers while mitigation maintains inspection controls for unavoidable risks.

**Logistics & Transport Risks (Risk Score: 6 – Medium)**

**Risk Description:**
Shipment delays or transport damage.

**Selected Strategy: Transfer + Mitigation**

**Actions Implemented:**

- Contractual penalties for late deliveries
- Use of insured logistics partners
- Packaging standards to reduce damage
- Delivery tracking systems

**Justification:**
Risk ownership is partially transferred to logistics providers while internal mitigation reduces damage likelihood.

**Environmental Storage Risks (Risk Score: 4 – Medium)**

**Risk Description:**
Material damage due to improper storage conditions.

**Selected Strategy: Mitigation**

**Actions Implemented:**

- Defined storage rules (temperature, stacking limits)
- Regular warehouse audits
- Training new workers on safe storage practices
- Clear labeling and zoning

**Why not Avoid?**
Storage is unavoidable so  mitigation is the only logical strategy.

**Cost & Budget Risks (Risk Score: 3 – Low)**

**Risk Description:**
Minor inventory holding cost fluctuations.

**Selected Strategy: Accept**

**Actions Implemented:**

- Continuous monitoring through ERP
- Monthly inventory cost review

**Justification:**
The cost of mitigation would exceed the potential impact, making acceptance the most efficient strategy.

## Evaluation of Risk Strategies

The applied risk strategies demonstrate a **mature and structured risk management approach** Instead of relying on a single response typess  strategies are selected based on:

- Risk severity
- Operational feasibility
- Cost–benefit balance
- Inventory system constraints

This flexibility aligns with best practices in engineering management and supports business continuity while minimizing unnecessary control costs.

Effective risk strategies transform the inventory department from a reactive function into a **proactive risk control system**. By selecting appropriate response strategies for each risk category, the department minimizes disruptions and  controls costs and  and protects operational performance.

## Risk Reduction strategic Recommendations.

Following the risk analysis performed on the project, a number of strategic suggestions are presented which could mitigate the occurrence and effects of the risks identified especially those associated with inventory management, dependency in the supply chain, and operational performance. These recommendations are based on long-term resiliency and control and prevention instead of short-term solutions.

**1. Enhancing Demand Forecasting to minimize the Inventory Risk.**

Amongst the most severe risks identified during the project is incorrect demand forecasting, which has a direct impact on the inventory levels, holding costs, as well as a threat of stockouts or overstocking. In order to counter this risk, the project ought to have in place a systematic forecasting strategy incorporating past trends of demand, seasonal trends and market indicators.

The implementation of rolling forecasts rather than periodic forecasts would enable the use of inventory decisions to be changed on a frequent basis with respect to the actual demand behavior. It decreases the risk of too much inventory and increases the service levels and demand responsiveness.

**2. Minimizing Supplier Dependency in Dual-Sourcing.**

High-impact risks identified were supplier-related risks, including delays, quality problems and lead-time variation. The use of one supplier makes it susceptible and may interfere with production and delivery processes.

In efforts to mitigate this risk, a dual-sourcing approach is advisable on the critical components. The project will be able to provide continuity of the supply through disruption in case of supply breakdown by qualifying at least one backup supplier. The strategy lowers the likelihood of supply disruptions and their consequences and raises supplier bargaining strength.

**3. Improving Inventory Visibility through Digital Systems.**

Little real time visibility of inventory status exposes the risk of human error, inaccurate stock records and delayed decision-making. To solve this, centralization of inventory management system combined with barcode tracking or RFID tracking is advised.

Real-time visibility enables early identification of any deviation including unusual consumption of stock or delayed replenishment as well as taking corrective measures before the risks occur. This is a direct contributor to the prevention of risks and not the reactive risk management.

**4. Introducing Safety Stock According to the Levels of Risk.**

Existing inventories are primarily pegged on minimum stock level and do not consider demand variability and supplier lead time contingencies. In order to minimize the effect of uncertainty, safety stock is to be determined by taking the variability of demand and lead time uncertainty of the important items.

Safety stock should be higher on the high-risk and high value parts and low risk items can be kept at leaner levels. This specific strategy minimises the chances of stockouts without a substantial rise in holding costs.

**5. Reenforcement of Quality Control to minimize Rework and Waste.**

Risks that were related to quality like flawed parts and damage caused during storage were cited to contribute to the costs and delay in operations. To eliminate these threats, delivery quality checks and the unified storage process must be strengthened.

Well-established storage policies, training of employees and periodic audits would help minimise damage and waste. This reduces the chances of risks associated with quality and enhances the general reliability of operations.

**6. Estimating Contingency and Escalation plans.**

In as much as the likelihood of risks reduces with the risk mitigation measures, some risks can still happen. As such, contingency and escalation plans are to be identified. These plans are to specify decision limits, position of action and response to actions in the event of supplier delays, demand overflow and system breakdowns.

Escalation paths are predefined to allow rapid response and minimize the effect of disruption to support business continuity in unanticipated circumstances.

**7. Constant Risk monitoring and review.**

Risk management cannot be taken as a single activity. Risk review process should be periodically introduced to re-evaluate the probability and impact levels on the basis of real performance information. This enables the project team to modify risk responses when the conditions vary and also makes the risk controls functional

# Cost Structure

## 1. Labor Cost Structure Monthly

### 1.1 Receiving Department Costs

| ROLE | QTY | AVG. SALARY (JOD) | TOTAL (JOD) |
|---|---|---|---|
| **RECEIVING SUPERVISOR** | 1 | 700 | 700 |
| **INBOUND LOGISTICS COORDINATOR** | 1 | 650 | 650 |
| **QUALITY INSPECTOR** | 1 | 600 | 600 |
| **DATA ENTRY CLERK** | 1 | 450 | 450 |
| **SUBTOTAL – RECEIVING** | | | **2,400** |

### 1.2 Storage Department Costs

| ROLE | QTY | AVG. SALARY (JOD) | TOTAL (JOD) |
|---|---|---|---|
| **WAREHOUSE MANAGER** | 1 | 800 | 800 |
| **WAREHOUSE SUPERVISORS** | 2 | 650 | 1,300 |
| **INVENTORY CLERK** | 1 | 500 | 500 |

| | | | |
|---|---|---|---|
| **MATERIAL HANDLERS** | 2 | 400 | 800 |
| **SUBTOTAL – STORAGE** | | | **3,400** |

**1.3 Dispatch Department Costs**

| ROLE | QTY | AVG. SALARY (JOD) | TOTAL (JOD) |
|---|---|---|---|
| **DISPATCH SUPERVISOR** | 1 | 700 | 700 |
| **PACKAGING & LABELING OFFICER** | 1 | 500 | 500 |
| **LOGISTICS COORDINATOR** | 1 | 650 | 650 |
| **DRIVER / LOADER** | 2 | 450 | 900 |
| **SUBTOTAL – DISPATCH** | | | **2,750** |

**1.4 Planning Department Costs**

| ROLE | QTY | AVG. SALARY (JOD) | TOTAL (JOD) |
|---|---|---|---|
| **INVENTORY PLANNER** | 1 | 800 | 800 |
| **PROCUREMENT COORDINATOR** | 1 | 700 | 700 |
| **BUDGET & COST ANALYST** | 1 | 750 | 750 |
| **ERP ANALYST** | 1 | 900 | 900 |
| **SUBTOTAL – PLANNING** | | | **3,150** |

**Total Monthly Labor Cost**

**2,400 + 3,400 + 2,750 + 3,150 = 11,700 JOD / month**

---

## 2. Operational & Overhead Costs (Monthly)

**2.1 Facility & Utilities**

| COST ITEM | MONTHLY COST (JOD) |
|---|---|

| | |
|---|---|
| **WAREHOUSE RENT / DEPRECIATION** | 2,000 |
| **ELECTRICITY & WATER** | 500 |
| **INTERNET & COMMUNICATION** | 150 |
| **SUBTOTAL** | **2,650** |

**Inventory Operations**

| COST ITEM | MONTHLY COST (JOD) |
|---|---|
| **PACKAGING MATERIALS** | 600 |
| **LABELS & PRINTING** | 200 |
| **HANDLING EQUIPMENT MAINTENANCE** | 400 |
| **PPE & SAFETY SUPPLIES** | 150 |
| **SUBTOTAL** | **1,350** |

**IT & Systems**

| COST ITEM | MONTHLY COST (JOD) |
|---|---|
| **ERP LICENSES & SUPPORT** | 700 |
| **BARCODE / SCANNER MAINTENANCE** | 200 |
| **IT SUPPORT SERVICES** | 300 |
| **SUBTOTAL** | **1,200** |

**Logistics & Transport**

| COST ITEM | MONTHLY COST (JOD) |
|---|---|
| **FUEL** | 800 |
| **VEHICLE MAINTENANCE** | 400 |
| **INSURANCE & REGISTRATION** | 250 |

| | |
|---|---|
| **TOTAL** | **1,450** |

**Total Monthly Overhead Cost**

**2,650 + 1,350 + 1,200 + 1,450 = 6,650 JOD**

## 3. Total Inventory Department Cost

**Monthly Cost**

- **Labor:** 11,700 JOD
- **Overhead:** 6,650 JOD

**Total Monthly Cost = 18,350 JOD**

**Annual Cost**

**18,350 × 12 = 220,200 JOD / year**

**quantified Variable Cost Breakdown**

**Table 1: Monthly Variable Inventory Costs**

| ***Variable Cost Category*** | ***Monthly Cost (JOD)*** |
|---:|---|
| *Packaging materials* | 600 |
| *Labels & printing* | 200 |
| *Inbound transportation* | 640 |
| *Outbound transportation* | 720 |
| *Fuel* | 800 |
| *Vehicle maintenance* | 400 |
| *Handling equipment maintenance* | 400 |
| *PPE & safety supplies* | 150 |
| *Overtime labor* | 175 |
| *Damage & waste* | 120 |
| *Forklift usage* | 90 |

| | |
|---|---|
| ***Total Variable Costs*** | **4,195 JOD** |

**Fixed Cost**

| *Fixed Cost Category* | *Monthly Cost (JOD)* |
|---|---|
| ***Base salaries (Inventory Dept.)*** | **11,700** |
| ***Warehouse rent / depreciation*** | **2,000** |
| ***ERP licenses & support*** | **700** |
| ***Internet & communication*** | **150** |
| ***Insurance & registration*** | **250** |
| ***Basic utilities (minimum load)*** | **300** |
| ***Security & safety contracts*** | **150** |
| ***Total Fixed Costs*** | **15,250 JOD** |

**Total Monthly Cost Comparison**

| COST TYPE | MONTHLY COST (JOD) | PERCENTAGE OF TOTAL |
|---|---|---|
| FIXED COSTS | 15,250 | 63.3% |
| VARIABLE COSTS | 4,195 | 36.7% |
| TOTAL INVENTORY COST | 19,345 JOD | 100% |

**Cost Behavior Analysis**

The comparison shows that fixed costs represent the larger share of total inventory costs, accounting for approximately 63% of monthly expenses. These costs provide operational stability but limit short-term flexibility.

Variable costs, while representing a smaller portion (37%), are highly sensitive to operational efficiency and risk management quality. Poor forecasting, supplier delays, or storage errors can significantly increase variable costs through overtime, emergency transport, and material damage.

From a managerial perspective:

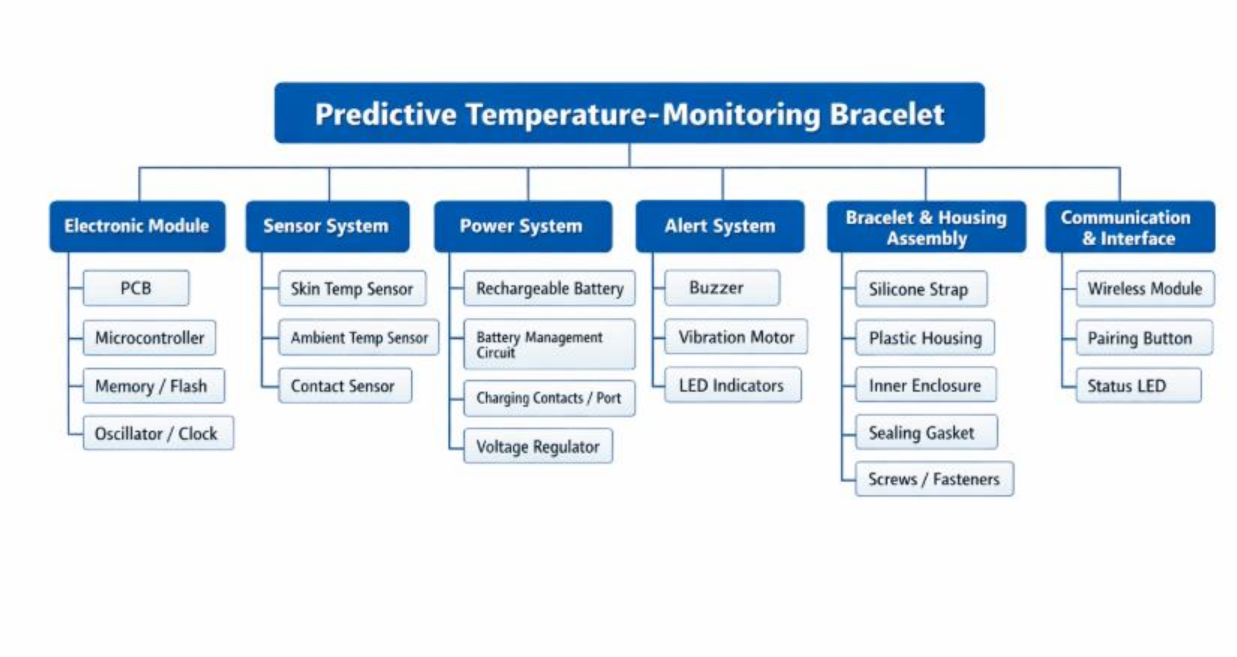
- Fixed costs define the baseline cost of operating the inventory system.
- Variable costs reflect how efficiently the inventory department is managed.

**Conclusion**

The comparison between fixed and variable costs highlights the strategic role of the Inventory Department in cost control. While fixed costs form the structural foundation of operations, variable costs represent the primary opportunity for financial optimization. A proactive, risk-aware inventory strategy therefore has a direct and measurable impact on total operational cost.

# Inventory Tools

## Multi level BOM



| ***Component* ID** | **Description** | **Type** | **Qty / Unit** | **Unit Cost (JOD)** | **Supplier (Local / Regional)** | **Current Stock** | **Reorder Frequency** |
|---|---|---|---|---|---|---|---|
| 1.00 | Predictive Temperature-Monitoring Bracelet | Finished Product | 1 | **1.5** | — | 50 | — |
| 1.01 | Electronic Module | Sub-assembly | 1 | 3.20 | — | 40 | — |
| 1.01.01 | PCB (Small, 2-layer) | Part | 1 | 0.80 | Jordan PCB Co. (Sahab) | 200 | 14 days |
| 1.01.02 | Microcontroller (MCU) | Part | 1 | 1.20 | Smart Electronics Amman | 150 | 21 days |
| 1.01.03 | Memory / Flash | Part | 1 | 0.40 | Smart Electronics Amman | 180 | 21 days |
| 1.01.04 | Oscillator / Clock | Part | 1 | 0.20 | Amman Electronics Market | 250 | 14 days |
| 1.02 | Sensor System | Sub-assembly | 1 | 2.10 | — | 45 | — |
| 1.02.01 | Skin Temperature Sensor | Part | 1 | 1.10 | RS Components Jordan | 120 | 30 days |
| 1.02.02 | Ambient Temperature Sensor | Part | 1 | 0.60 | RS Components Jordan | 130 | 30 days |
| 1.02.03 | Placement / Contact Sensor | Part | 1 | 0.40 | Amman Electronics Market | 160 | 21 days |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1.03 | Power System | Sub-assembly | 1 | 2.50 | — | 40 | — |
| 1.03.01 | Li-Po Battery (Small) | Part | 1 | 1.60 | Battery House Jordan (Marka) | 100 | 30 days |
| 1.03.02 | Battery Protection Circuit | Part | 1 | 0.40 | Smart Electronics Amman | 150 | 21 days |
| 1.03.03 | Charging Contacts | Part | 1 | 0.20 | Amman Electronics Market | 200 | 14 days |
| 1.03.04 | Voltage Regulator | Part | 1 | 0.30 | RS Components Jordan | 180 | 21 days |
| 1.04 | Alert System | Sub-assembly | 1 | 1.20 | — | 50 | — |
| 1.04.01 | Buzzer | Part | 1 | 0.30 | Amman Electronics Market | 300 | 14 days |
| 1.04.02 | Vibration Motor | Part | 1 | 0.60 | Amman Electronics Market | 250 | 14 days |
| 1.04.03 | LED Indicator | Part | 2 | 0.15 | Amman Electronics Market | 500 | 14 days |
| 1.05 | Bracelet & Housing Assembly | Sub-assembly | 1 | 2.30 | — | 60 | — |
| 1.05.01 | Silicone Strap | Part | 1 | 0.90 | Plastics Industries Co. (Zarqa) | 200 | 21 days |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1.05.02 | Plastic Outer Housing | Part | 1 | 0.80 | Middle East Plastic (Sahab) | 180 | 21 days |
| 1.05.03 | Inner Enclosure | Part | 1 | 0.30 | Middle East Plastic (Sahab) | 180 | 21 days |
| 1.05.04 | Sealing Gasket | Part | 1 | 0.20 | Rubber Tech Jordan | 300 | 14 days |
| 1.05.05 | Screws / Fasteners | Part | 4 | 0.10 | Jordan Fasteners Co. | 500 | 30 days |
| 1.06 | Communication & Interface Module | Sub-assembly | 1 | 1.20 | — | 40 | — |
| 1.06.01 | Wireless Module (Short-range) | Part | 1 | 0.90 | Smart Electronics Amman | 100 | 30 days |
| 1.06.02 | Pairing Button | Part | 1 | 0.15 | Amman Electronics Market | 300 | 14 days |
| 1.06.03 | Status LED | Part | 1 | 0.15 | Amman Electronics Market | 400 | 14 days |

## current stock and suppliers

| **Item ID** | **Component** | **Supplier** | **Current Stock (Units)** | **Minimum Stock** | **Reorder Point (ROP)** | **Lead Time (Days)** | **Stock Status** |
|---|---|---|---|---|---|---|---|
| 1.01.01 | PCB (2-layer) | Jordan PCB Co. (Sahab) | 200 | 80 | 120 | 14 | OK |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1.01.02 | Microcontroller (MCU) | Smart Electronics Amman | 150 | 70 | 100 | 21 | OK |
| 1.02.01 | Skin Temperature Sensor | RS Components Jordan | 120 | 60 | 90 | 30 | Monitor |
| 1.02.02 | Ambient Temperature Sensor | RS Components Jordan | 130 | 60 | 90 | 30 | OK |
| 1.03.01 | Li-Po Battery | Battery House Jordan (Marka) | 100 | 50 | 75 | 30 | Monitor |
| 1.04.01 | Buzzer | Amman Electronics Market | 300 | 120 | 180 | 14 | OK |
| 1.04.02 | Vibration Motor | Amman Electronics Market | 250 | 120 | 180 | 14 | OK |
| 1.05.01 | Silicone Strap | Plastics Industries Co. (Zarqa) | 200 | 100 | 150 | 21 | OK |
| 1.05.02 | Plastic Housing | Middle East Plastic (Sahab) | 180 | 90 | 130 | 21 | OK |
| 1.06.01 | Wireless Module | Smart Electronics Amman | 100 | 50 | 80 | 30 | Monitor |

**BOM – Warehouse Capacity Calculations**

Total available warehouse capacity = **500 storage units**

The warehouse space is allocated based on BOM level importance and value:

- Parts: **50%**
- Sub-assemblies: **30%**
- Finished products: **20%**

**Parts Storage Calculation**

$$500 \times 0.5 = 250 \text{ units for parts}$$

Number of different parts = **25**

$$250 \div 25 = 10 \text{ units per part}$$

**Result:**
Each individual part can be stored up to **10 units**.

**Sub-Assemblies Storage Calculation**

$$500 \times 0.3 = 150 \text{ units for sub-assemblies}$$

Number of sub-assemblies = **5**

$$150 \div 5 = 30 \text{ units per sub-assembly}$$

**Result:**
Each sub-assembly can be stored up to **30 units**.

**Finished Products Storage Calculation**

$$500 \times 0.2 = 100 \text{ finished product units}$$

**Result:**
The warehouse can store **100 finished product units**.

**Reorder Frequency Logic**

- **Finished products** have the **shortest reorder cycle** due to high value and direct customer demand.

- **Sub-assemblies** follow with a **moderate reorder frequency** to support assembly flow.
- **Parts** have the **longest reorder cycle**, considering production and manufacturing lead times.

Supplier deliveries should be coordinated to arrive in closely aligned time windows to prevent production

# Part III

# persuasion and Negotiation

## 1. Difference Between Persuasion and Negotiation

Negotiation and persuasion are closely related concepts, but they serve different purposes within managerial and organizational contexts.

**Negotiation** In simplest terms, negotiation is a discussion between two or more disputants who are trying to work out a solution to their problem. This interpersonal or inter-group process can occur at a personal level, as well as at a corporate or international level. Negotiations typically take place because the parties wish to create something new that neither could do on his or her own, or to resolve a problem or dispute between them The parties acknowledge that there is some conflict of interest between them and think they can use some form of influence to get a better deal rather than simply taking what the other side will voluntarily give them. They prefer to search for agreement rather than fight openly, give or break off contact.


When parties negotiate, they usually expect give and take. While they have interlocking goals that they cannot accomplish independently, they usually do not want or need exactly the same thing. This interdependence can be either win-lose or win-win  in nature, and the type of negotiation that is appropriate will vary accordingly. The disputants will either attempt to force the other side to comply with their demands, to modify the opposing position and move toward compromise, or to invent a solution that meets the objectives of all sides. The nature of their interdependence will have a major impact on the nature of their relationship, the way negotiations are conducted, and the outcomes of these negotiations.

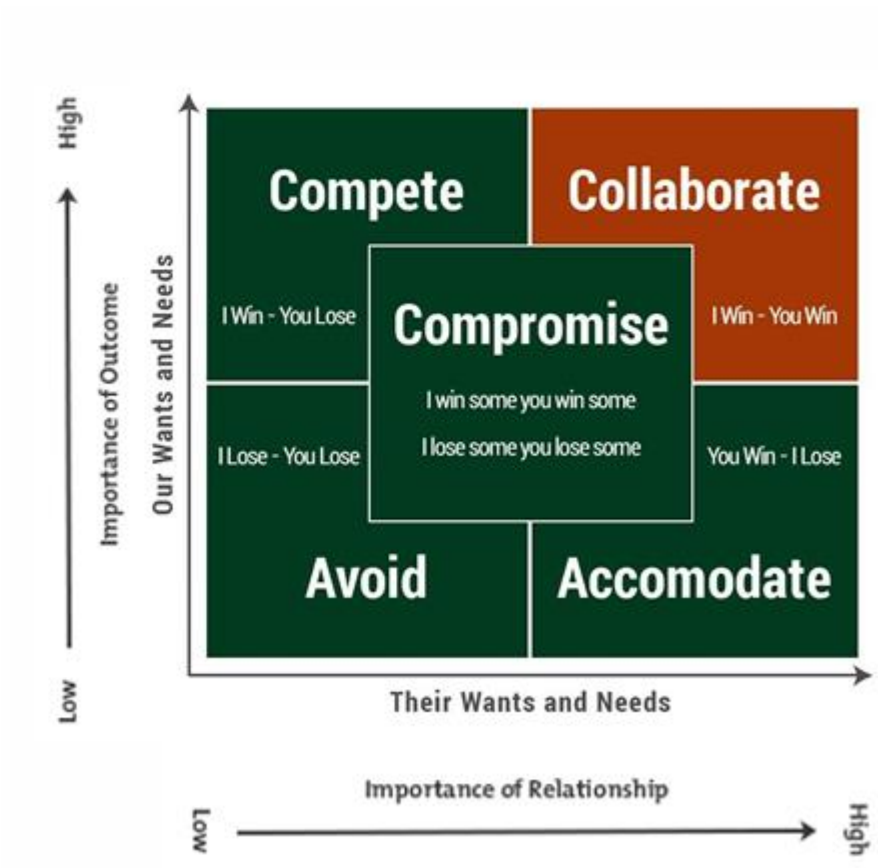
**Persuasion** is a symbolic process in which a communicator attempts to convince others to change their beliefs, attitudes, or behaviors regarding a specific issue through the transmission of a message in an atmosphere of free choice. Unlike coercion, which uses force or threats, true persuasion relies on the audience's willing acceptance of an idea based on the speaker's credibility emotional resonance and logical reasoning In modern psychology, this process is often explained by how deeply an individual processes information through a critical, central route or a more superficial, peripheral route resulting in either lasting or temporary shifts in perspective and  it is a goal-oriented form of communication that seeks to align the listener's viewpoint with the speaker's intent while respecting the listener's autonomy. In the context of persuasive writing, the **OREO style** is a popular mnemonic framework used to structure an argument so that it is clear, convincing, and easy to follow. It organizes a persuasive message into four distinct layers:

- **O – Opinion:** Start by clearly stating your position or what you want the audience to believe
- **R – Reason:** Provide a logical "why" to support your opinion, appealing to the audience's intellect
- **E – Explanation:** Back up your reason with specific facts, data, or personal anecdotes to make the argument
- **O – Opinion (Restated):** Wrap up by reiterating your original stance in a new way to leave a lasting impression

By using this structure, persuasion becomes more than just a feeling; it becomes a well-organized sandwich of logic and evidence that is hard for an audience to reject.

## Negotiation Style Matrix

The negotiation style matrix presented in the lecture visualizes negotiation behavior based on two dimensions:



- **Importance of the outcome**
- **Importance of the relationship**

Based on these dimensions, five negotiation styles are identified:

**1.Avoiding (Lose–Lose)**

This style is used when the issue is not important and the relationship is also not important. People avoid the discussion and do not try to solve the problem. It is suitable for very small or temporary issues. However, it should not be used for important problems, because ignoring them may cause bigger issues later.

**2. Accommodating (Lose–Win)**

In this style, the relationship is more important than the outcome. One party accepts the other's demands, often to maintain harmony or because of limited power. This approach is inappropriate when it leads to unfair outcomes or long-term dissatisfaction.

**3. Competitive (Win–Lose)**

The competitive style focuses on winning the negotiation, even if the relationship is damaged. It is useful in urgent situations or when one side is sure they are right and there is no time for discussion. However, this style can harm long-term relationships and should be avoided when cooperation is important.

**4. Compromise (Win Some – Lose Some)**

This style is used when time is limited and both sides have similar power. Each side gives up something to reach a quick agreement. It is practical, but the final solution may not be the best one, especially when the issue is very important.

**5. Collaborating (Win–Win)**

The collaborating style aims to satisfy both sides and keep a strong relationship. It is best used when the issue is important and the relationship matters. This style requires time and open communication, but it helps build trust, transparency, and long-term cooperation.

## 3. Practical Example Using the Negotiation Style Matrix

In the Inventory Department, a negotiation situation occurred with a long-term supplier after receiving a shipment with **poor quality materials** and **extended delivery lead time**. This issue had not occurred before, and the organization had maintained a strong, long-term relationship with the supplier.

Based on the negotiation style matrix, the **Collaborating (Win–Win)** style was selected. The outcome was important because poor material quality directly affected production efficiency and product reliability. At the same time, the relationship was also critical, as the supplier had been reliable in the past and represented a strategic partner.

During the negotiation process, the preparation phase focused on collecting quality reports, delivery records, and historical performance data. In the discussion phase, both parties openly exchanged concerns, allowing the supplier to explain the root causes of the quality and delivery issues. In the proposal and bargaining phases, joint solutions were explored, such as improving inspection procedures, adjusting lead time commitments, and introducing corrective action plans.

Persuasion techniques were applied by presenting factual evidence, emphasizing mutual long-term benefits, and using professional communication rather than authority or pressure. This collaborative approach resulted in improved quality controls, reduced lead time, and strengthened trust between both parties, demonstrating the effectiveness of the win–win negotiation style.

# Leadership

## Leadership Theories

In engineering management, leadership plays a critical role in guiding teams through complex tasks there is four main theories are often highlighted

### Trait Theory

trait Theory suggests that effective leaders possess inherent personal characteristics such as integrity and  confidence also  empathy and  assertiveness, and strong decision-making abilityin  the context of inventory management these traits are essential because the manager is responsible for coordinating multiple functions, resolving conflicts and making quick decisions that directly affect cost service level, and operational continuity.

As an Inventory Manager integrity is crucial when handling stock discrepancies or reporting inventory errors Empathy is especial important when managing new employees who are unfamiliar with warehouse systems and procedures. Assertiveness is required when enforcing safety rules, inventory discipline, and operational standards. while these traits support leadership effectiveness, Trait Theory alone is limited because it does not explain how leaders adapt to different operational situations.

### Behavioral Theory

Behavioral Theory focuses on what leaders do rather than who they are. In warehouse operations, leadership behavior directly impact employee performance, as well error ratesand  and workflow efficiency Common leadership behaviors include autocratic democratic and laissez-faire approaches.

In the Inventory Department autocratic behavior may be necessary during critical receiving or dispatch operations where safety and accuracy are non negotiable Democratic behavior becomes good when discussing process improvements, layout changes or Kaizen courses  with experienced staff. Laissez-faire leadership can be applied selectively when working with high skill employees who can operate independent with minimal supervision This theory is practical because it make observable leadership actions that can be adjusted and improved over time.

## Situational/Contingency Theory

Situational and Contingency Theory states that there is no single best leadership style. Instead, effective leadership depends on the situation task urgency and team capability This theory is particularly relevant in inventory operations where conditions change daily.

And  when experienced staff are available and operations are stable, a participative leadership style works well. However, during peak seasons, system issues, or when onboarding new workers, a more directive approach is required. As an Inventory Manager, shifting between leadership styles ensures that operational goals are met without sacrificing team morale or accuracy.

## transactional and Transformational Leadership

Transactional leadership focuses on structure, rules, performance monitoring, and corrective actions. This style is common in inventory environments where KPIs such as stock accuracy, picking errors in time dispatch must be controll Employees is  rewarded for compliance and performance and corrected when deviations occur.

Transformational leadership on the other hand, focuses on motivation, engagement, and long term improvement In the Inventory department this can be seen when encouraging staff to participate in continuous improvement initiatives, suggesting layout improvements, or reducing handling waste. A balanced use of both styles ensures operational discipline while promoting employee engagement and ownership.

## Leadership Style Matrix

The leadership style matrix compares two factors:

- Task orientation: how much focus is placed on getting the job done.
- Relationship orientation: how much focus is placed on the team and people.

| Style | Task-Oriented | People-Oriented | Description |
|---|---|---|---|
| Directive | High | Low | Focused on efficiency and structure |
| Supportive | Low | High | Builds trust  listens to team |
| Participative | Medium | High | Involves team in decision-making |
| Achievement-Oriented | High | High | Pushes team to achieve challenging goals |

## Discussion of the Leadership Style Matrix

The leadership style matrix is useful because it helps managers adapt their approach based on what the team needs like if a project is falling behind and people need structure a directive style makes sense. But if the team is stressed or unmotivated a supportive approach might be better to rebuild trust and morale.


*Figure 3 Leadership Style Matrix*

the Leadership Style Matrix is a pragmatic and adaptable model that assists managers to realize how they can modify their leadership behavior according to two important dimensions, including task orientation and relationship orientation. Instead of selling a single best style of leadership, the matrix stresses that the best leadership style is concerned with the circumstances, the capacity of the team, and the urgency of the task.

Significance of the Task and Relationship Orientation.

Task orientation is an aspect that indicates the level of attention a leader has on planning, structure, deadlines, and targets of performance. The aspect is critical in the field of engineering and operation where accuracy, safety, and efficiency have a direct impact on the results. Relationship orientation, in its turn, shows how the leader directs his/her attention towards communication, trust, motivation, and the well-being of the team. Any negligence in this dimension in a people intensive environment may result into low morale, resistance and low productivity.

It is in the combination of these two dimensions that the leaders can respond to various issues that are affecting their operations.

Ladership Styles Analysis in the Matrix.

Directive Leadership (High Task Low Relationship)

The directive type of style focuses on structure, control and efficiency. It is also highly applicable when they have urgent tasks, when the process is new, or when the members of the team are inexperienced. This style comes in handy in the process of engineering projects at the crucial stages like failure of the system, safety accidents, or deadlines that necessitate precise guidelines and swift decision-making.

Nevertheless, excess application of this style may decrease the level of engagement and creativity in the team, particularly where it is used in circumstances where it is not critical.

Supportive Leadership (Low Relationship High Task)

Supportive leadership is a people-oriented leadership that is not process-oriented. It works best when there is stress in teams, de-motivation and inter-personal problems. The leader restores trust and psychological safety by listening actively and being concerned about the team members.

This type of style can be useful in an engineering setting once the periods of high pressure are over or when mistakes are made, since the focus on learning can be promoted instead of blame. Nevertheless, the application of supportive leadership alone can result in a lack of direction in case performance expectations are not specified.

Participative Leadership (Medium Task -High Relationship)

The participative approach is a moderate structure and collaboration. The leaders engage employees in decision-making process, appreciate their contributions, and promote collective accountability. This is the most efficient in cases where the time can be spent in iscussion and in cases where the members of the team are specialists in the field.

Participative leadership may result in a better decision in the case of engineering projects, where technical issues can be improved with multiple views. It also enhances commitment since people in the team will feel to be in charge of the end product. The primary weakness is that it can slow down decision making under time sensitive circumstances.

Achievement-Oriented Leadership (High Task -High Relationship)

This is a style that incorporates high expectations and high levels of support. Leaders have tough targets to accomplish as they demonstrate belief in the strength of the team to accomplish. It is best performed under the condition that the teams are talented, motivated, and independent workers.

Achievement oriented leadership drives teams beyond the minimum standards and promotes innovation and excellence in the advanced engineering or continuous improvement environment. But when it is used in a team that lacks experience, it can cause pressure and stress instead of motivation.

Real Life Use of Leadership Style Matrix.

Its flexibility is the major advantage of the leadership style matrix. It enables leaders to deliberately change their style instead of actually sticking to a leadership identity. Engineering projects in the real world are dynamic- competence within the team varies, risk also varies and time lines are also subject to change. The leader that is aware of the matrix can change.

As an illustration, a project which begins with a directive approach because of strict deadlines can later shift to a participative approach when the processes become stable. Likewise, in high workload or error correction seasons, a supportive style might be required, and afterwards, the performance-oriented leadership style can be adopted.

## Example in the Inventory  Department

During the onboarding of new warehouse employees, several storage errors occurred, including incorrect bin placement and mislabeling of inventory. These errors increased stock discrepancies and slowed down picking operations.

Using the Leadership Style Matrix, the situation was identified as **high programmability but low autonomy**, since warehouse procedures were clearly defined, but the new employees lacked experience. As a result, an autocratic and directive leadership style was applied temporarily. Clear instructions were given tasks were closely supervis and daily feedback sessions were conducted.

As employees gained experience and confidence, leadership shifted to a more participative style. Team members were encouraged to ask questions, suggest improvements, and take responsibility for specific storage zones. This gradual shift increased inventory accuracy, reduced errors, and improved team confidence.

This example demonstrates how leadership flexibility guided by the leadership matrix, leads to better operational outcomes

# Coaching and Mentoring

The development methodologies that are necessary in the operations like inventory and warehouse management include the use of coaching and mentoring. Inventory is generally sensitive, and the accuracy, safety and timing of inventory seriously affect the overall performance of the business hence the need of the employees in the inventory situation to be constantly guided, trained and motivated. Coaching and mentoring are similar terms, but they have a great amount of difference in goals, organization, and time perspective.

The choice of the style of operation in the inventory department directly relates to the accuracy of stock, availability of the material and efficiency in the operations.

## Differentiation between Coaching and Mentoring.

The Coaching in the Inventory Context.

Coaching is a short-term and performance-oriented development model. Coaching is often used in inventory activities in cases where an employee shows deficiencies in performing tasks including wrong stock registration, mishandling of materials or neglecting of storage processes.

Through a coach, an employee works closely to:

Determine areas of performance shortage

Give straight forward directions and examples.

Establish short term improvement objectives.

Track the progress and give feedback.

The coaching most appropriately applies to inventory settings where errors affect costs, safety or manufacturing directly.

Mentoring within the Inventory Environment.

The mentoring approach is concerned with long-term professional development, as opposed to addressing short-term task-specific problems. When it comes to the inventory department, mentoring is instrumental to helping the employees get a wider perspective on the concepts of supply chain, inventory planning and decision making.

A mentor typically:

Shares have experience in inventory management and warehouse control.

Develops the career (e.g. storekeeper to inventory controller) of a guide.

Develops trust and ability to solve.

Creates future leaders in the department.

Mentoring relations are unofficial, relationship based, and developed over time.

Inventory- Based Comparison Coaching vs Mentoring.

| Aspect | Coaching | Mentoring |
|---|---|---|
| **Primary Focus** | Correcting performance and skill gaps | Long-term development and career growth |
| **Time Frame** | Short-term | Long-term |
| **Structure** | Highly structured and task-oriented | Flexible and relationship-based |
| **Inventory Application** | Fixing stock errors, improving accuracy | Developing inventory planning and leadership skills |
| **Advantage** | Fast performance improvement | Sustainable employee growth |
| **Limitation** | Limited to specific tasks | Slower visible results |
| **Best Used When** | Immediate operational issues exist | Preparing employees for future roles |

Inventory Management Inventory management Skill Will Matrix.

Skill Will Matrix is a potent management instrument to be used in the assessment of inventory staff in the light of:

Skill: Technical skill to carry out inventory work properly.

Will: A desire and a determination to do the job.

This matrix enables the inventory managers to implement the best leadership and development strategy to each category of employees.

High Will -High Skill (Delegate)

These types of employees are experienced inventory personnel, those who are conversant with the storage regulations, the ERP and safety measures. They are trustworthy and enterprising.

Management Approach:

Minimal supervision delegation. These workers will be able to manage the number of cycles, audit of stock and resolving discrepancies on their own.

High Skill – Low Will (Engage)

Such employees are highly knowledgeable in regard to their inventory but lack the motivation which is usually as a result of the repetitive nature of their jobs, lack of incentive or an assigned role.

Management Approach:

Re-engagement should be done by increasing responsibility, participation in improvement activities and recognising the knowledge they have.

Low Skill – High Will (Coach)

This group is usually composed of fresh employees in the warehouse who are well motivated but have no experience. They are also liable to making mistakes like improper storae places or labeling mistakes.

Management Approach:

Coaching is essential. Managers develop competence through provision of hands-on training, standard operating procedures (SOPs), shadowing and frequent feedback.

Low Skill – Low Will (Direct)

Workers in this quadrant have a weak capability and low motivation, which will be of high risk to the accuracy and safety of inventory.

Management Approach:

Directive leadership including instructions, over-checking as well as performance control. Reassignment or improvement plans might need to be provided in case performance is not improved.

SkillWill Matrix Inventory Department.

The task of a new warehouse worker employed in the inventory department was to receive and store electronic parts. Although the employee was highly motivated, he/she kept on putting materials in the wrong places resulting in picking mistakes and delays in production.

According to the Skill-Will Matrix, this employee fell in the Low Skill-High Will category. Coaching approach was used which involved:

Practical training of storage layout and labeling requirements.

Progressive instructions in receiving tasks.

Feedback on a daily basis in the first working period.

This translated to a great deal less storage errors, an improved inventory accuracy and a trustworthy team player with the employee.

Inventory Operations Coaching/Mentoring Critique.

Even though coaching and mentoring are efficient developmental instruments, they are limited in the operational environment. Coaching is a process that needs constant supervision and managerial time, which might prove difficult when there is peak inventory. Mentoring is useful in the long-term perspective of building capability but it does not automatically correct the mistakes in operations.

Hence, inventory managers should be able to select a suitable approach according to the level of skills possessed by the employees, their motivation, and urgency of operations. Integrated coaching to achieve short term performance enhancement and mentoring to achieve long term growth is a sure way of stabilizing operations and growing the workforce in the long term.

## Part IIII

## Social Responsibility of Engineers

**1. Public Safety and Public Interest**

Engineers have a primary responsibility to place public safety and public interest above all other considerations. Any engineering decision to save human life health 0wellbeing must be careful evaluat and control Engineers is expecte to reduce risks and prevent unsafe practices before they cause harm.

**2. Awareness of Social Impact**

Engineering solutions often have long-term effects on society. Engineers must consider both the positive and negative consequences of technology and operational decisions. This includes understand how processes systems or designs do affect employees and the wider community.

**3. Sustainability and Environmental Responsibility**

Engineers are responsible for minimizing environmental impact by promoting sustainable practices This includes efficient use of materials and reducing waste lowering energy consumption and and supporting environmentally responsible operations.

**4. Ethical Decision Making and Speaking Up**

Social responsibility requires engineers to speak up when they observe unsafe, unethical, or harmful decisions even if do is so is uncomfortable Remai silent in the presence of harmful practices contradicts professional responsibility.

**5. Equality and Diversity and Inclusion**

Engineers should promote fairness also respect and equal opportunities in the workplace. Social responsible engineering environments value diversity and ensure that decisions are not influenced by discrimination or bias

**6. Knowledge Sharing and Community Contribution**

Engineers have a responsibility to share knowledge and expert for the benefit of society. This can include mentoring so training and public awareness, or contributing professional skills to community-oriented initiatives.

**Application of Social Responsibility in Inventory Department**

**Public Safety in Inventory Operations**

In the Inventory Department, public safety is a daily responsibility rather than a theoretical concept. As the Inventory Manager, I encountered a situation where a decision was proposed to store materials in an unsafe manner due to space limitations and operational pressure. This storage method posed a risk to workers due to unstable stacking and insufficient safety clearance.

Despite time pressure I stopped this decision and enforced proper storage standards other solutions were implemented including reorganizing storage zones and reducing congestion This action reflect the engineer social responsibility to prioritize safety over speed or shortterm .

**Reducing Waste and Environmental Impact**

Another key responsibility applied in the Inventory Department is sustainability. Inventory mismanagement often leads to excessive waste through damaged materials, expired items, or unnecessary overstocking. By improving stock organization, applying FIFO principles, and enhancing inventory visibility, material damage and waste were reduced.

These actions not only lowered operational costs but also reduced environmental impact by minimizing  and reordering This demonstrates how responsible inventory management supports both business efficiency and environmental protection.

**Ethical Decision Making and Accountability**

Social responsibility requires accountability for operational decisions. When storage or handling errors occur, it is ethically important to acknowledge mistakes rather than conceal them. In our department, errors are documented, analyzed, and corrected to prevent recurrence.

This approach creates a culture of responsibility and trust, where employees feel encouraged to report issues rather than hide them Ethical accountability strengthens operational reliability and team confidence.

**Fair Treatment and Workforce Responsibility**

Managing a warehouse involves continuous interaction with workers from different backgrounds and experience levels Social responsibility requires treat all employees fairly and ensuring that safety rules and workload distribution apply equally to everyone.

By providing clear instructions fair task allocation, and supportive supervision, the Inventory Department promotes an inclusive and respectful work environment. Thisreduces conflict so  increases morale, and improves operational performance.

**Speaking Up for Social Responsibility**

One of the most critical aspects of social responsibility is the willingness to speak up against unsafe or unethical practices. As an Inventory Manager stopping unsafe storage decisions and correcting wasteful practices reflects the engineer's duty to protect people and resources.

This aligns with the principle that engineers don't  only follow instructions but also question decisions that may cause  Responsible leadership requires moral courage, not just technical competence.

**Knowledge Sharing and Continuous Improvement**

Social responsibility is also reflected in knowledge sharing and continuous improvement. Training new employees, explaining safety standards, and encouraging best practices ensure that responsible behavior becomes part of daily operations.

By supporting learning and improvement, the Inventory Department contributes positively to both organizational culture and operational sustainability.

# Ethics

## Core Ethical Principles

1. **Public Safety First**
   the Inventory Department, prioritizing public and employee safety means:

   - **Safe Storage:** Organizing materials and stacking pallets correctly to prevent collapses.
   - **Right Handling:** Use the right procedures to move goods without injury or damage.
   - **Compliance:** Follow safety regulations to prevent workplace accidents.

2. **Honesty and Integrity**

   - Honest **Reporting:** Record exact stock levels, even when there are shortages or damages.
   - No **Cover up :** Report discrepancies immediately rather than hiding them to avoid blame.

• Operational **Trust:** Accurate data is essential for reliable financial reporting, ordering, and planning.

## 3. Sustainability

**Minimize environmental impact**

- Reduce Waste: Avoid overstock and reuse packaging.
- Eco Efficiency: Properly disposing of expired goods and streamlining transport to save energy

## 4. Accountability

**Take ownership of our work:**

- Fix Mistakes: Correct mislabeling or stock errors immediately.
- Improve: Use errors as lessons to prevent them from happening again.

## 5. Competence

**Work only where you are trained:**

- Skill Matching: Only operate machinery or software if certified.
- Continuous Learning: Ensure all staff are trained before starting new tasks.

## 6. Confidentiality

**Protect sensitive data:**

- **S**ecure Info: Keep supplier prices are also stock levels and forecasts private.
- Prevent Leaks: Ensure unauthorized people cannot access internal inventory systems.

## 7. Fairness

**Treat everyone equally:**

- Equal Opportunity: Base work assignments and promotions on merit not bias.
- Respect: Foster a workplace where all team members are treated with dignity.

## 8. Transparency

**Communicate clearly and accurately:**

- Clear Records: Log every stock movement in the ERP system immediately.

- Audit-Ready: Ensure documentation is easy for other departments to follow

**9. Intellectual Property**

**Respect others ideas :**

- Legal Use: Do not copy a supplier unique packaging designs or barcoding systems without permission.

## Application in Our Project and inventory Department

In our business, the Inventory Department play a critical role in ensuring ethical engineering practices across the supply chain. Public safety is maintained by enforcing strict safety rules during receiving, storage, and dispatch operations. Honesty and transparency are reflected in accurate inventory reporting and open communication with planning and procurement teams.

Sustainability principle are applied by optimize stock levels to avoid waste and unnecessary reorder. Accountability and competence are reinforced through defined responsibilities and and structured training programs for new employees and continuous performance monitoring. Confidentiality is maintained by restricting access to sensitive inventory and supplier data within the ERP system.

Fair treatment of employees encourages teamwork and reduces operational risks caused by disengagement or poor morale. Respect for intellectual property and ethical sourcing strengthens long term relationships with suppliers as well as a clear ethical reporting culture ensures that any unethical behavior is addressed promptly and responsibly.

# CPD

## Inventory Management course

**course Title : Inventory Management**
University of California
Instructor: Paul Jan
Platform: Coursera

course link *:* https://www.coursera.org/learn/inventory-management#outcomes

**Why I Select This Course**

this course because inventory management is a critical function in my department. My role involves controlling stock levels, managing reorder decisions, minimizing waste and damage, and balancing service levels with cost efficiency.

The course directly supports my professional responsibilities by focusing on:

- Inventory optimization in uncertain environments
- Demand variability and safety stock decisions
- Cost trade-offs between holding, ordering, and shortage costs
- Practical inventory analysis using real data

This aligns strongly with the challenges I face in managing electronic components, sub-assemblies, and finished products within limited warehouse capacity.

**What I Learned from the Course**

After completing multiple sessions from the course, I gained the following key knowledge and skills:

**Inventory Cost Structure**

I developed a deeper understanding of inventory-related costs, including:

- **Holding costs**: storage, space utilization, damage, and obsolescence
- **Ordering costs**: procurement processing, supplier coordination, and transportation
- **Shortage costs**: production delays, emergency orders, and loss of service level

This helped me clearly differentiate between **fixed costs** (warehouse rent, ERP systems) and **variable costs** (handling per unit, packaging, damage, urgent procurement).

**Reorder Point (ROP) and Safety Stock Logic**

The course clarified how reorder points should be calculated based on:

- Average demand
- Lead time
- Demand variability
- Desired service level

I learned how poor ROP decisions can directly increase operational risk, which I later applied when reviewing components with long lead times and frequent shortages in my inventory.

**Data-Driven Inventory Decisions**

The course emphasized using historical data rather than intuition alone. I learned how inventory analytics can support:

- Monitoring slow-moving and fast-moving items
- Identifying overstock vs understock situations
- Improving forecast accuracy

This reinforced the importance of ERP accuracy and cycle counting in my department.

**Application to My Project and Department (Do Phase)**

I applied the course learning directly in my warehouse operations:

- I reviewed existing stock levels against reorder points and lead times.
- I identified items with **high holding cost and low turnover** and adjusted replenishment frequency.
- I supported decisions to **reduce excess stock**, minimizing damage and waste.
- I improved communication with procurement by justifying reorder decisions using data instead of assumptions.

This directly improved inventory visibility and supported safer and more cost-effective storage practices.

**Reflection on Learning**

Reflecting on this CPD activity, I realized that many past inventory issues were not caused by suppliers alone, but by insufficient data analysis and weak inventory planning logic. The course helped me move from a reactive approach to a more structured, analytical mindset.

It also increased my confidence in explaining inventory decisions to senior management using quantitative reasoning rather than subjective judgment.

**Recording and Evidence (Record Phase)**

This CPD activity is documented through:

- Coursera course enrollment
- Completion of multiple learning sessions
- Application of learned concepts in my inventory report and calculations
- Reflection included in this professional development section

This ensures my CPD record demonstrates continuous learning and practical application, not just attendance.

**CPD Cycle Alignment**

This activity follows the CPD cycle presented in the course material:

| CPD Stage | Application in My Case |
|---|---|
| **Plan** | Identified need to improve inventory decision-making |
| **Do** | Completed Inventory Management course sessions |
| **Reflect** | Evaluated how learning improved my warehouse practices |
| **Record** | Documented learning and application in this report |

**Future CPD Actions**

Based on this experience, I plan to:

- Continue developing skills in inventory analytics and demand forecasting
- Enroll in advanced supply chain and data analysis courses
- Apply predictive techniques to further reduce risk and cost in inventory operations

## Inventory Analytics Course

**Course Title:** Inventory Analytics

**Instructor:** [Yao Zhao]
**Provider:** Rutgers University
**Course Link:** https://www.coursera.org/learn/inventoryanalytics#outcomes

**Overview of the Course**

This course focuses on understanding inventory from an analytical and financial perspective. It explains how inventory decisions directly affect company performance, cash flow, and operational efficiency. The course introduces analytical tools used to identify inventory problems, classify inventory items, and improve decision-making using real business cases.

As a manager responsible for inventory and operations, this course was highly relevant to my role and departmental responsibilities.

**Key Knowledge and Skills Gained**

From this course, I developed the following skills and understanding:

- Understanding how inventory impacts **financial performance**, profitability, and cash cycle
- Identifying early signs of inventory problems such as overstocking, slow-moving items, and stock imbalance
- Learning how to **classify inventory** high-value vs low-value items) instead of managing all items equally
- Linking inventory control to **materials management, supply chain efficiency, and operational risk**

**Application to My Department and Project**

In my inventory department, I applied the concepts learned in this course in a practical and realistic way:

- I became more aware of how excess inventory ties up capital and increases storage and handling costs
- I improved prioritization of inventory items by focusing more on **critical and high-impact components**
- I used analytical reasoning to justify inventory decisions when communicating with senior management
- The course supported better alignment between inventory decisions and overall business performance

This helped me move from operational inventory control to **strategic inventory management**.

**Reflection (CPD – Reflect Stage)**

This course enhanced my ability to view inventory not only as a storage function but as a strategic business asset. It improved my understanding of how inventory decisions influence cost structure, service level, and financial outcomes. The analytical approach gained from this course strengthened my role as a manager and supported better, data-driven decisions in my department.

**CPD Cycle Summary for This Course**

| ***CPD Stage*** | ***Description*** |
|---|---|
| ***Plan*** | Improve inventory decision-making and financial awareness |
| ***Do*** | Completed Inventory Analytics course sessions |
| ***Record*** | Documented skills gained and analytical tools learned |
| ***Reflect*** | Applied inventory analytics to improve prioritization and reduce waste |

## Demand Planning Course

**Course title:** Demand Planning in RStudio: Create Demand Forecast
**Platform:** Coursera
**Instructor:** Moses Gummadi
**Field:** Demand Planning / Forecasting / Supply Chain Analytics

Course link : https://www.coursera.org/projects/demand-forecasting#outcomes

**Course Overview**

This guided project focuses on demand forecasting using real data and practical tools. Unlike theoretical courses, this course is fully hands-on and teaches how to analyze historical demand data, identify trends and seasonality, and build quantitative demand forecasts that support production planning and inventory decisions.

The course is highly relevant to my role as a manager responsible for inventory and planning decisions, as demand uncertainty directly affects stock levels, reorder points, and inventory risk.

**Key Learning Outcomes**

During this guided project, I learned how to:

- Analyze historical demand data

- Identify **trend and seasonality patterns**
- Categorize products based on demand behavior
- Build demand forecasting models using RStudio
- Generate demand forecasts for multiple future periods
- Evaluate forecast accuracy and model reliability

**Skills Practiced**

This course strengthened the following practical skills:

- **Demand Forecasting**
- **Production Planning**
- **Supply Chain Planning**
- **Data Analysis**
- **Trend and Seasonality Analysis**
- **Use of RStudio for analytics**
- **Decision-making based on data, not assumptions**

**Application to Inventory Department**

The knowledge gained from this course was directly applicable to my project and daily work in the inventory department.

In my project:

- Demand forecasting helped me **estimate future material requirements** instead of relying on static minimum stock levels.
- I used the forecasting logic to support **inventory planning and reorder decisions**, especially for components with long lead times.
- Understanding seasonality allowed me to anticipate **demand fluctuations**, reducing the risk of overstock and stockouts.
- Forecast outputs supported better alignment between **inventory levels, warehouse capacity, and production planning**.

The course enabled me to move from reactive inventory management to **proactive planning based on data**.

**Reflection Using the CPD Cycle**

**Plan**

I identified a need to improve my ability to forecast demand accurately to support inventory and production decisions.

**Do**

I completed this guided project and applied demand forecasting techniques using real datasets and analytical tools.

**Record**

I documented forecasting methods, model outputs, and how they influence inventory and planning decisions.

**Reflect**

I realized that accurate demand forecasting is a key driver for:

- Lower inventory costs
- Better service levels
- Reduced operational risk
- More confident managerial decisions

This CPD activity significantly improved my professional competence in **data-driven supply chain planning**.

**Conclusion**

The  Demand Planning in RStudio: Create Demand Forecast course provided practical,relevant skills that directly support my role as a manager. It strengthened my ability to link **demand data with inventory and production planning**, making my decisions more accurate, defensible, and aligned with business objectives.

# Industry 4.0 Trends and Their Application in a Predictive Health Monitoring Project

## Introduction

Modern engineering industries  increase transformation driven by digitalization, automation and data driven decision making This transformation, commonly an its name **Industry 4.0** represents a shift from traditional manufacturing systems to  smart system and intelligent operations As highlighted in recent industrial research including large scale implementations in advanced manufacturing companies, Industry 4.0 integrates technologies such as the Internet of Things (IoT) artificial intelligence  big data analytics, and cyber-physical systems to increase  efficiency and reliability and  and responsiveness

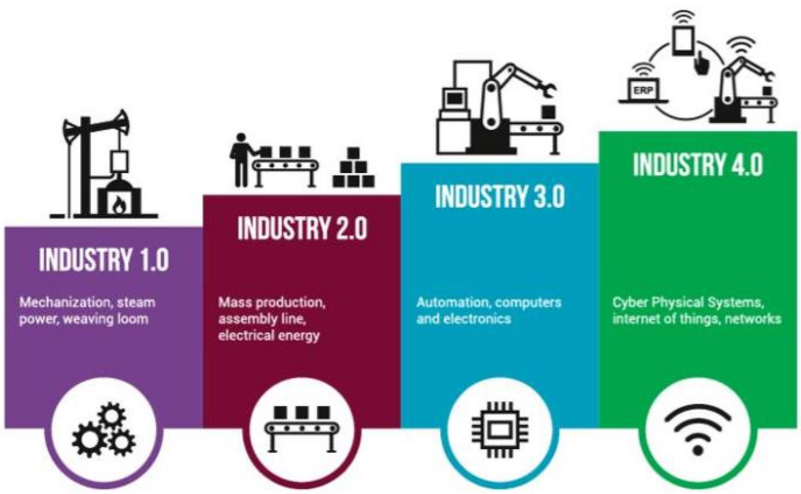



*Figure 4 Industry Timeline*

significant trends in modern engineering is the **predictive and proactive systems** rather than reactive ones. Engineering processes are no longer limited to monitoring current conditions but are increasingly focused on forecasting future events, reducing uncertainty, and preventing failures before they occur. This trend is evident in areas such as predictive maintenance and smart quality control and  and intelligent supply chain management, where real-time data and analytics enable early detection of risks and performance deviations.

These trends are directly relevant to my project  **Predictive Temperature Monitoring Bracelet for Children** The product itself reflects Industry 4.0 principles by combining continuous sense  predictive analytics and autonomous decision making to enhance safety and reliability and the Inventory Department supporting this product can benefit from Industry 4.0 trends through smarter inventory planning realtime stock visibility, predictive demand forecasting and improved coordination across receiving storage and dispatch operations bt applied inventory management with modern industrial trends the project aims to achieve higher reliability faster response times and improved overall operational performance.

## Key Industry 4.0 Trends in Modern Engineering

Based on recent industrial research and large scale in advanced manufacturing environments many l key trends define Industry 4.0 and modern engineering systems The following trends are particle relevant to my project and its operational steps.

### IoT Smart Manufacturing

The Internet of Things enables physical devices machine and and sensors to collect and exchange data in real time in modern engineering environments IoT supports smart manufacturing by providing continuous visibility into system conditions and performance. This allows engineers and operators to detect deviations early and respond quickly.

In the context of this project IoT principles are reflected in the continuous temperature and ambient sensors embedded in the bracelet These sensors provide real-time data that supports predictive analysis and early detection of From an operational perspective, IoT concepts can also be applied in inventory operations through smart tracking of sensitive components such as sensors and batteries.

### AI & Predictive Maintenance

Artificial intelligence (AI) enables systems to analyze data patterns and predict future outcomes. In engineering application AI is widely used for predictive maintenance, quality prediction, and fault prevention. Rather than reacting to failures after they occur, AI-driven systems aim to prevent failures before they happen

This trend aligns strongly with the predictive nature of the temperature monitoring bracelet. The project uses predictive temperature trend analysis to forecast potential fever events instead of relying solely on threshold based alerts like within the Inventory Department, AI-based forecasting can be used in the future to predict demand fluctuations and supplier risks, improving reliability and reducing emergency interventions.

## Big Data in Supply Chain & Inventory

Big data analytics refers to the processing and analysis of large volumes of operational data to support better decision-making. In supply chain and inventory management, big data enables more accurate demand forecasting, inventory optimization, and risk identification.

For this project, big data principles support inventory planning decisions such as reorder points, safety stock levels, and supplier performance evaluation. By analyzing historical consumption, lead time variability, and quality records, the Inventory Department can reduce both overstocking and stockouts. This ensures stable production and timely availability of critical components for the bracelet.


*Figure 5 Big data in logistics*

## Digital Twins and  Simulation

Digital twins are virtual representations of physical systems that allow simulation and analysis before real world implementation In engineering environment digital twins are used to test process change and  optimize layouts and evaluate risks without stopped the ongoing operations.

In the future  digital twin concepts can be applied to simulate inventory layouts, material flow, and dispatch processes supporting the bracelet project. This allows the organization to evaluate improvement scenarios, reduce implementation risks, and make data-driven decisions before physical changes are introduced.

## Human-Centered Automation

Industry 4.0 don't aim to replace human workers entirely instead modern engineering increasing focase on **human-centered automation**, where digital tools support human decision making rather than eliminate it This combines lean principles with digital technologie to improve efficiency also reduce errors also and enhance employee engagement.

In inventory and warehouse operations human centered automation is particularly important. ERP system handheld scanners and digital dashboards support staff by reduc manual work and providing accurate information while human judgment remains essential for quality checks and exception handling, and continuous improvement activities such as Kaizen.

## Application Industry 4.0 Trends to the Project

The Predictive Temperature Monitoring Bracelet for Children project can applie Industry 4.0 principles at both the **product level** and the **operational level** The project is not only a smart product but also part of a broader smart operational system supported by inventory and engineering processes.

| ***Industry 4.0 Trend*** | ***Application in the Project*** | ***Value Created*** |
|---|---|---|
| *IoT & Smart Manufacturing* | Continuous temperature and ambient sensors | Real-time monitoring and higher accuracy |
| *AI & Predictive Analytics* | Fever trend prediction algorithm | Early risk detection and prevention |
| *Big Data Analytics* | Inventory planning and demand forecasting | Reduced stockouts and excess inventory |
| *Digital Twin & Simulation* | Future process and layout simulations | Lower implementation risk |
| *Human-centered automation* | ERP-supported inventory decisions | Efficiency with human control |

## Impact on Inventory Department and Operations

Industry 4.0 trends significantly influence how the Inventory Department supports the project. High product reliability requires accurate inventory control, traceability, and timely availability of components. Data-driven inventory planning improves coordination between receiving, storage, and dispatch operation

| Engineering Requirement | Inventory Application | Operational Benefit |
|---|---|---|
| **Reliability** | Controlled storage of sensors and batteries | Reduced defect risk |
| **Predictive planning** | Data-based reorder points | Improved availability |
| **Traceability** | ERP batch tracking | Quality and compliance |
| **Speed** | Optimized dispatch processes | Shorter lead times |
| **Risk reduction** | Analytics-based decision-making | Operational continuity |

## Future Implementation and Strategic Value

While some digital elements are already implemented through ERP systems and basic automationthe project provides a foundation for future Industry 4.0 expansion advanced analytics IoT enabled inventory tracking and simulation tools can further improve performance and scalability.

From a strategic perspective ans adopting Industry 4.0 trends enables the organization to move from reactive operations toward predictive and resilient systems. This alignment between product design and operational processes strengthens safety and  efficiency, and long-term competitiveness Industry 4.0 represents a fundamental shift in modern engineering, emphasizing intelligence connectivity  and predictive capabilities. The Predictive Temperature Monitoring Bracelet for Children reflects these trends through its design and operational support structure apply Industry 4.0 principles to both engineering and inventory operations the project achieves higher reliability  better risk control  and improved operational performance  This approach positions the project as a future ready solution aligned with global industrial best practices.

# Meeting Agendas and MOMs

## Meeting Agenda #1

**Date:** 24/11/2025
**Time:** 8:00 – 9:00 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Meeting Objectives**

- Go over the progress of the first submission
- Assign the remaining tasks for the first submission

**Agenda Items**

**Previous Progress:**

- Literature review
- Legal requirements
- Organizational structure and explanation
- Risk overview
- Report template

**New Tasks:**

- Business Model Canvas (BMC)
- Fishbone diagrams and contingency plans
- Supply chain diagram

- Total Quality Management (TQM)

**Minutes of Meeting #1**

**Date:** 24/11/2025
**Time:** 8:00 – 9:00 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Points Discussed in the Meeting**

- Went over the progress of the first submission
- Most sections were drafted, with BMC, fishbone diagrams, supply chain, and TQM still pending
- Literature review was ready but needed summarizing and reformatting
- Legal requirements and organizational structure were mostly completed
- The report template was finalized
- Assigned the remaining tasks for the first submission:
    - BMC
    - Supply chain diagram
    - Fishbone diagrams and TQM

**Action Items – Meeting #1**

| ACTION ITEMS | OWNER(S) | DEADLINE | STATUS |
|---|---|---|---|
| Create report template | Maria | 26/11/2025 | Completed |
| Organizational chart | Diab | 26/11/2025 | Completed |
| Legal requirements and budgeting | Omar | 27/11/2025 | Completed |
| Summarize and reformat literature review | Abdalla | 27/11/2025 | Completed |
| Business Model Canvas | Maria | 28/11/2025 | Completed |
| Supply chain diagram and TQM | Omar, Diab | 28/11/2025 | Completed |
| Fishbone diagrams | All members | 28/11/2025 | Completed |

## Meeting Agenda #2

**Date:** 15/12/2025
**Time:** 6:50 – 8:15 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Meeting Objectives**

- Go over second submission progress
- Re-check Business Model Canvas (BMC)

**Agenda Items**

**Previous Progress:**

- All of first submission and received feedback

**New Tasks:**

- Risk assessment
- Cost structure per department
- Department-specific tools
- Break-even analysis
- Life Cycle Assessment (LCA)

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**Minutes of Meeting #2**

**Date:** 15/12/2025
**Time:** 6:50 – 8:15 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Points Discussed in the Meeting**

- Reviewed progress of the second submission
- Each member reported progress:
    - Maria: Risk assessment, manufacturing risks
    - Diab: Design risks and technical justification

    - Omar: Cost structure and inventory risks
    - Abdalla: Marketing risks and break-even inputs
- Re-checked BMC and confirmed alignment across all departments

**Action Items – Meeting #2**

| ACTION ITEMS | OWNER(S) | DEADLINE | STATUS |
|---|---|---|---|
| Risk assessment section | Maria | 18/12/2025 | Completed |
| Legal requirements and budgeting review | Omar | 18/12/2025 | Completed |
| Summarize and update risk register | Abdalla | 19/12/2025 | Completed |
| Business Model Canvas update | Maria | 19/12/2025 | Completed |
| Supply chain risks and TQM alignment | Omar, Diab | 19/12/2025 | Completed |
| Break-even analysis | All members | 20/12/2025 | Completed |
| Fix SCM alignment | Omar | 20/12/2025 | Completed |

## Meeting Agenda #3

**Date:** 30/12/2025
**Time:** 7:00 – 7:50 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Meeting Objectives**

- Finalize tasks in the second submission
- Discuss third submission tasks

**Agenda Items**

**Second Submission:**

- Review cost structure and break-even analysis
- Share opinions on individual work

**Third Submission:**

- Business process model (Camunda)
- Individual leadership and CPD sections

**Minutes of Meeting #3**

**Date:** 30/12/2025
**Time:** 7:00 – 7:50 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Points Discussed in the Meeting**

- Finalized second submission tasks
- Reviewed RBS and cost structure for each department
- Ensured break-even analysis accuracy
- Confirmed all risk analysis was completed
- Discussed third submission tasks:
    - Ensured each member understands their BPMN role

- Clarified individual leadership, persuasion, and CPD sections

**Action Items – Meeting #3**

| ACTION ITEMS | OWNER(S) | DEADLINE | STATUS |
|---|---|---|---|
| Risk section final review | Maria | 02/01/2026 | Completed |
| Cost structure validation | Omar | 02/01/2026 | Completed |
| Supply chain logic review | Diab | 02/01/2026 | Completed |
| Business process model (Camunda) | All members | 03/01/2026 | Completed |
| Leadership & CPD drafting | All members | 04/01/2026 | Completed |

# Meeting Agenda #4

**Date:** 12/01/2026
**Time:** 8:00 – 9:30 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Meeting Objectives**

- Go over the entire submission tasks, focusing on common tasks

**Agenda Items**

**Final Submission:**

- Business Model Canvas
- Cost structure
- Legal requirements
- Break-even analysis
- Supply chain and logistics

**Minutes of Meeting #4**

**Date:** 12/01/2026
**Time:** 8:00 – 9:30 pm
**Location:** Online
**Attendees:** Maria, Diab, Omar, Abdalla

**Points Discussed in the Meeting**

- Reviewed final submission sections
- BMC: confirmed communication of value proposition
- Cost structure: verified fixed and variable costs
- Legal: confirmed capital and registration details
- Break-even analysis: validated calculations
- SCM and logistics: ensured final version consistency

**Action Items – Meeting #4**

| ACTION ITEMS | OWNER(S) | DEADLINE | STATUS |
|---|---|---|---|
| Final report formatting | Maria | 15/01/2026 | Completed |
| SCM and logistics final version | Diab, Omar | 15/01/2026 | Completed |
| Cost structure final check | Omar | 15/01/2026 | Completed |
| Break-even final validation | Abdalla | 15/01/2026 | Completed |
| Final submission | All members | 17/01/2026 | In progress |

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